

Scholars High School Physics 1: Mechanics (4926)

Andrew Sayler

Fri

Sep 11, 2026 - Mar 12, 2027

7:30 PM - 9:00 PM ET (4:30 PM - 6:00 PM PT)

Overview

Week 2 (Sep 18) Class Transcript - Acceleration



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aninconle 2026-09-18 19:30:37

Scholars High School Physics 1: Mechanics Lesson 2: Acceleration

aninconle 2026-09-18 19:30:39

Hi 😊

LCLARK5691 2026-09-18 19:30:55

Hello!

Thorongil 2026-09-18 19:30:55

Hi!

hw_zasf.xAx 2026-09-18 19:30:55

hello

syl314 2026-09-18 19:30:55

Hi!

AN29 2026-09-18 19:30:55

hi!

Brightgiraffe 2026-09-18 19:30:55

hello

Zip-a-Dee-Doo-Dah 2026-09-18 19:30:55

Hello! 😊

kabir_sahil 2026-09-18 19:31:06

Hello

Ash_Ketchup13 2026-09-18 19:31:06

hello

vmene 2026-09-18 19:31:06

Hi!

mocha_x 2026-09-18 19:31:06

hi

flybirdeee_ 2026-09-18 19:31:06

hii

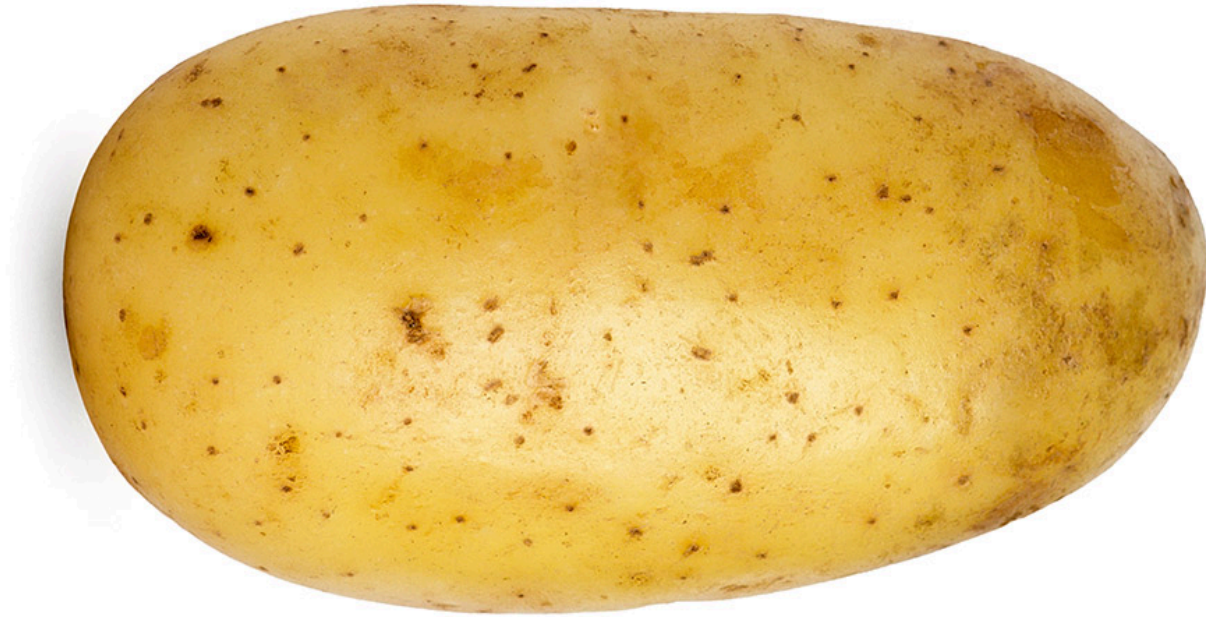
aahelic13 2026-09-18 19:31:06

hi

MEBat 2026-09-18 19:31:06

Hello, everyone! 😊

mocha_x 2026-09-18 19:31:08



aninconle 2026-09-18 19:31:14

And a good potato to you too.

aninconle 2026-09-18 19:31:18

Speeding Up

aninconle 2026-09-18 19:31:24

What's this?



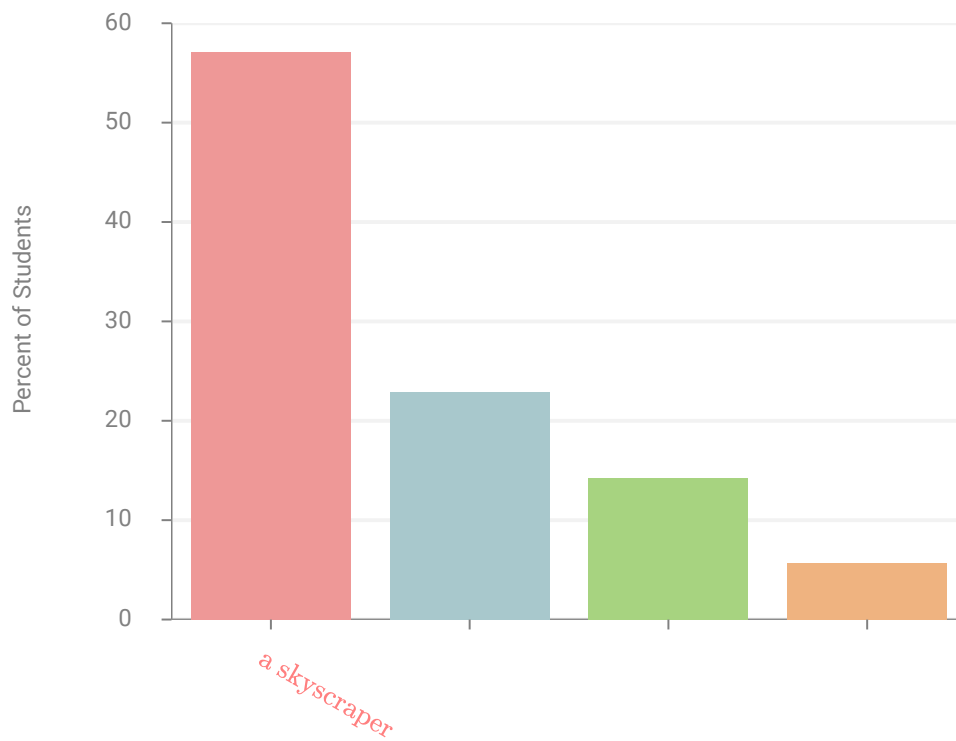
a skyscraper

a dragonfly doing yoga, but the head, wings, and legs fell off

a pretty cool cloud, but part of it is blocked by something

the twisty slide at my local playground

aninconle 2026-09-18 19:32:10



aninconle 2026-09-18 19:32:15

That's Shanghai Tower. It's 632 m tall. It has a fast elevator with a display that shows you how high you are.

aninconle 2026-09-18 19:32:17



aninconle 2026-09-18 19:32:29

The elevator rises 10 m in 5.2 s. How long will it take to rise 630 m?

mocha_x 2026-09-18 19:34:20

327.6s

vmene 2026-09-18 19:34:20

327.6 seconds

Nebula233 2026-09-18 19:34:20

327.6 seconds

Brightgiraffe 2026-09-18 19:34:20

327.6 seconds

Ash_Ketchup13 2026-09-18 19:34:20

327.6 seconds

aahelic13 2026-09-18 19:34:20

327.6 seconds

Diamond_Zombie67 2026-09-18 19:34:20

327.6s, 6 minute

mathisfun2026 2026-09-18 19:34:20

327.6s

Potatomiaomi 2026-09-18 19:34:20

327.6 sec

Canada1334 2026-09-18 19:34:20

Or 328 s

nickypicky39084 2026-09-18 19:34:20

327.6

aninconle 2026-09-18 19:34:24

We could assume height is proportional to time. The elevator has gone $\frac{1}{63}$ of the way up, so it needs 63 times as long as 5.2 s, or 328 s, to go all the way up.

aninconle 2026-09-18 19:34:31

What's wrong with this approach?

VeryDragon2013 2026-09-18 19:36:20

we dont know if its constant speed!

syl314 2026-09-18 19:36:20

It speeds up as it goes

apollo2000 2026-09-18 19:36:20

it doesn't account for acceleration or change in speed

VeryDragon2013 2026-09-18 19:36:20

it could be accelerating!!

Potatomiaomi 2026-09-18 19:36:20

it will accelerate?

Thorongil 2026-09-18 19:36:20

the elevator is slow at the beginning, speeds up, then has to slow down to a stop

mathisfun2026 2026-09-18 19:36:20

the elavator's speed isn't constant

Canada1334 2026-09-18 19:36:20

The elevator never changes its speed

thekitsune 2026-09-18 19:36:20

the it takes to accelerate

Exponent11 2026-09-18 19:36:20

acceleration

WildOctopus 2026-09-18 19:36:20

the elevator speeds up

Seifr 2026-09-18 19:36:20

We don't know that the elevator's velocity is constant.

LCLARK5691 2026-09-18 19:36:20

the velocity might not be constant

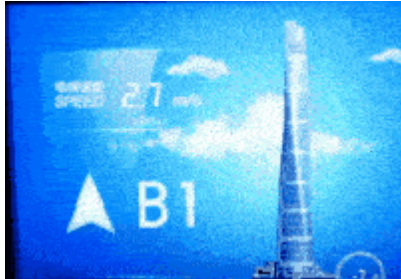
aninconle 2026-09-18 19:36:22

Distance and time are only proportional at constant speed, but the elevator is getting faster over time.

aninconle 2026-09-18 19:36:24

Here's a zoomed out view of that display. (It's sped up to save time.) You can see the speedometer changing in the top left.

aninconle 2026-09-18 19:36:25



aninconle 2026-09-18 19:36:33

The elevator can reach the top in just 55 s.

aninconle 2026-09-18 19:36:36

Model the position of the elevator over time.

aninconle 2026-09-18 19:36:44

After speeding up, the elevator goes at a constant speed, then slows down.

aninconle 2026-09-18 19:36:45



aninconle 2026-09-18 19:36:50

This is a dot diagram with arrows added, called a *motion diagram*. The arrows tell us the direction of the motion. What does the length of an arrow represent?

kabir_sahil 2026-09-18 19:39:02

How much it moves in a given time period

Thorongil 2026-09-18 19:39:02

the distance the elevator traveled in a certain time frame

mathisfun2026 2026-09-18 19:39:02

how far it can travel in a fixed time

flybirdeee_ 2026-09-18 19:39:02

how much the elevator traveled during one second

Ash_Ketchup13 2026-09-18 19:39:02

The length of the arrow represents the distance traveled by the elevator in a specific time frame.

vmene 2026-09-18 19:39:02

what distance it covered in a constant amount of time

aninconle 2026-09-18 19:39:07

The arrows are the displacement during each time interval. They are also proportional to the velocity. Long arrows mean the elevator is going fast.

aninconle 2026-09-18 19:39:09

Develop the concept of acceleration to describe changing velocity.

aninconle 2026-09-18 19:39:10

When the velocity of an object is changing, the object has *acceleration*.

aninconle 2026-09-18 19:39:12

Acceleration is how quickly the velocity is changing, also called the *rate of change* of the velocity.

aninconle 2026-09-18 19:39:15

In everyday life, people typically use "acceleration" for increasing speed and "deceleration" for decreasing speed. In physics, we don't use "deceleration."

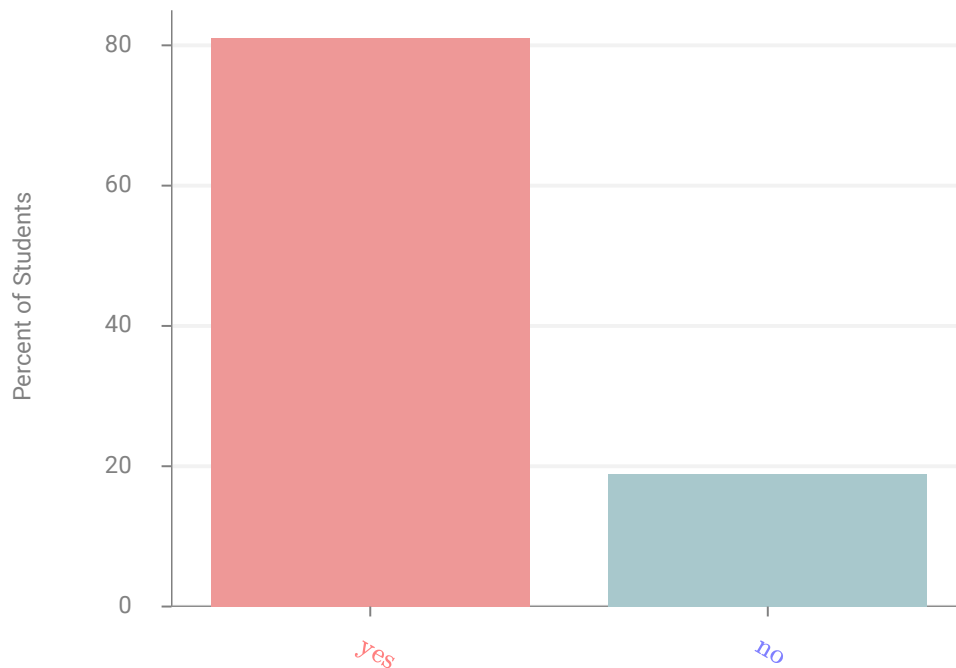
aninconle 2026-09-18 19:39:17

When the elevator is nearing the top, it slows down. Is this an acceleration?

yes

no

aninconle 2026-09-18 19:40:36



aninconle 2026-09-18 19:40:40

Yes it is—the velocity is changing.

aninconle 2026-09-18 19:40:41

Let's use y as a coordinate, since the elevator moves vertically. Velocity is the rate of change of y ,

$$v_{y;\text{avg}} = \frac{\Delta y}{\Delta t}.$$

aninconle 2026-09-18 19:40:43

Drag the elements to create an equation for average acceleration.

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

flybirdeee_ 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

aahelic13 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

agoel2010 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

Thorongil 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

VeryDragon2013 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

apollo2000 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

Seifr 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

Brightgiraffe 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

Canada1334 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

Oceanix 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

syl314 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

Potatomiaomi 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

Grace_174 2026-09-18 19:41:49

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}$$

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We can create a new equation for acceleration by replacing Δy in the velocity equation with Δv_y

$$a_{y;\text{avg}} = \frac{\Delta v_y}{\Delta t}.$$

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Remember that subscript $_y$ means that the variable includes direction. a_y can be either positive or negative, depending on the direction of acceleration. a is the magnitude of acceleration, $a = |a_y|$.

aninconle 2026-09-18 19:41:59

What's the average acceleration of the elevator over the entire ride?

aninconle 2026-09-18 19:42:02

What do you get for this problem?

AN29 2026-09-18 19:43:37

0m/s^2

Thorongil 2026-09-18 19:43:37

0

VeryDragon2013 2026-09-18 19:43:37

wait 0 because the velocity goes from 0 back to 0 right?

Ash_Ketchup13 2026-09-18 19:43:37

0

aninconle 2026-09-18 19:43:39

The elevator has zero velocity at both the beginning and end of the ride. Δv_y is

$$\begin{aligned}\Delta v_y &= v_{y;\text{final}} - v_{y;\text{initial}} \\ &= 0 - 0 \\ &= 0.\end{aligned}$$

aninconle 2026-09-18 19:43:55

The average acceleration over the whole ride was

$$\begin{aligned}a_{y;\text{avg}} &= \frac{\Delta v_y}{\Delta t} \\ &= \frac{0}{\Delta t} \\ &= 0.\end{aligned}$$

aninconle 2026-09-18 19:44:02

The elevator clearly *did* have acceleration during the trip though. How can it be that we average that acceleration over the entire trip, including that parts where acceleration was not zero, and still get zero in the end?

syl314 2026-09-18 19:46:29

There was positive and negative

Brightgiraffe 2026-09-18 19:46:29

positive and negative acceleration cancel out?

Alexmo2022 2026-09-18 19:46:29

acceleration sped up and slowed down

Zip-a-Dee-Doo-Dah 2026-09-18 19:46:29

The positive and negative acceleration cancel out.

mathisfun2026 2026-09-18 19:46:29

the acceleration and negative acceleration(deceleration) cancel out

Zip-a-Dee-Doo-Dah 2026-09-18 19:46:29

The positive and negative acceleration cancel out. (From the perspective of the average)

vmene 2026-09-18 19:46:29

it goes positive and negative

aninconle 2026-09-18 19:46:31

The acceleration was positive sometimes and negative others.

aninconle 2026-09-18 19:46:33

We can understand positive and negative acceleration if we draw some *graphs*.

aninconle 2026-09-18 19:46:34

Graphs of Motion

aninconle 2026-09-18 19:46:35

Make a velocity versus time graph for the elevator's rise.

aninconle 2026-09-18 19:46:37

Here's our motion diagram again, this time with the arrows labeled:

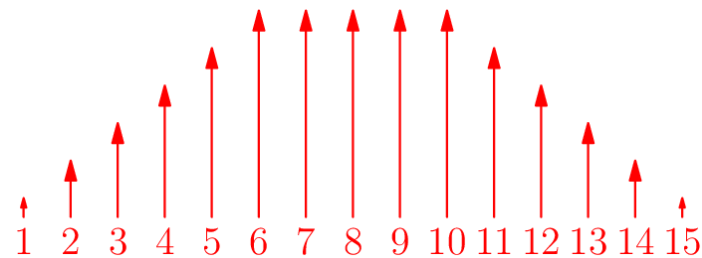
aninconle 2026-09-18 19:46:38



aninconle 2026-09-18 19:46:41

Let's take all the arrows and put them next to each other in a horizontal row.

aninconle 2026-09-18 19:46:42



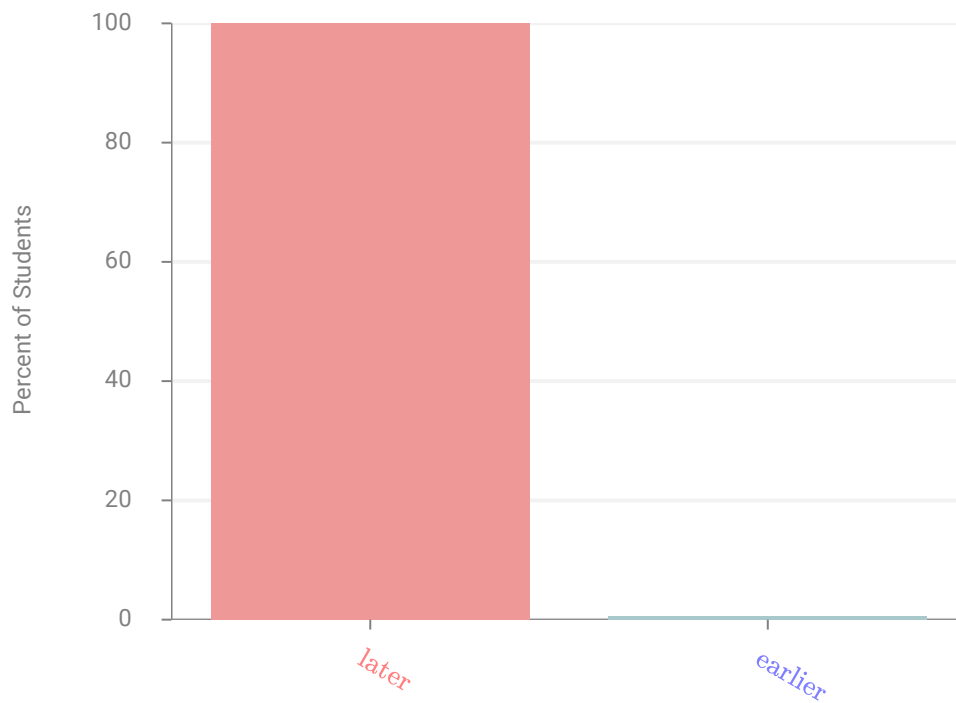
aninconle 2026-09-18 19:46:44

The further to the right, the ... the time we're looking at.

later

earlier

aninconle 2026-09-18 19:47:38



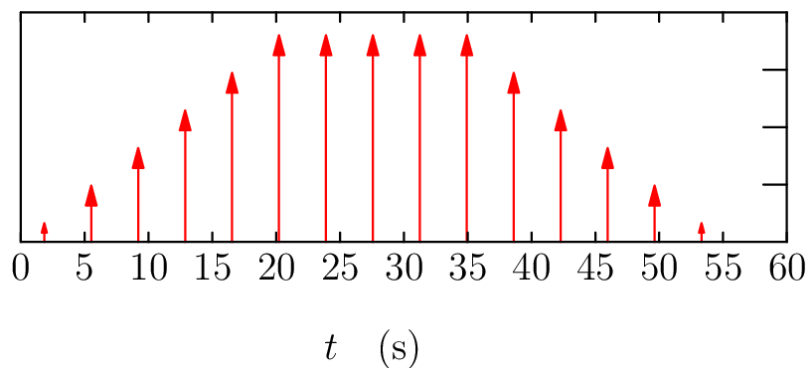
aninconle 2026-09-18 19:47:44

The numbers increase to the right, which means later times.

aninconle 2026-09-18 19:47:47

Let's add axes to our graph. The horizontal one will indicate time.

aninconle 2026-09-18 19:47:49



aninconle 2026-09-18 19:47:52

Most physicists use axes on all sides of the figure, so we've done that here.

aninconle 2026-09-18 19:47:55

What should we label the vertical axis?

Thorongil 2026-09-18 19:49:24

v for velocity

agoel2010 2026-09-18 19:49:24

velocity

Grace_174 2026-09-18 19:49:24

velocity

WildOctopus 2026-09-18 19:49:24

velocity

AN29 2026-09-18 19:49:24

the velocity

apollo2000 2026-09-18 19:49:24

Velocity

kabir_sahil 2026-09-18 19:49:24

Displacement

thekitsune 2026-09-18 19:49:24

velocity

nickypicky39084 2026-09-18 19:49:24

velocity

Grace_174 2026-09-18 19:49:24

v (m/s)

thekitsune 2026-09-18 19:49:24

(m/s)

TatuShark 2026-09-18 19:49:24

velocity

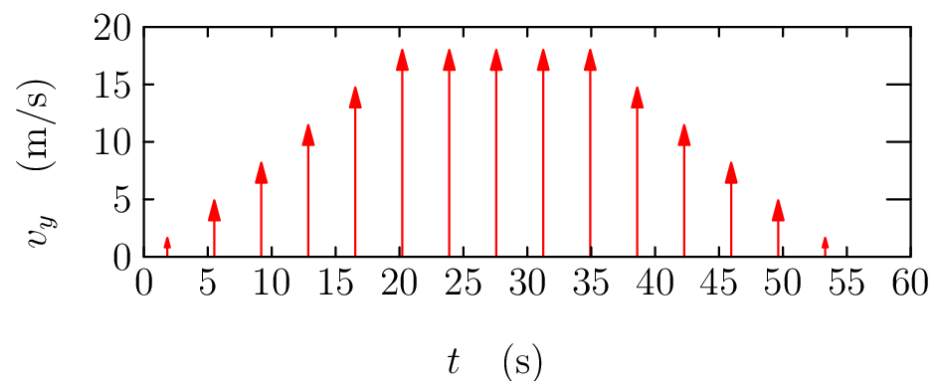
aahelic13 2026-09-18 19:49:24

velocity

aninconle 2026-09-18 19:49:25

It's the velocity, v_y .

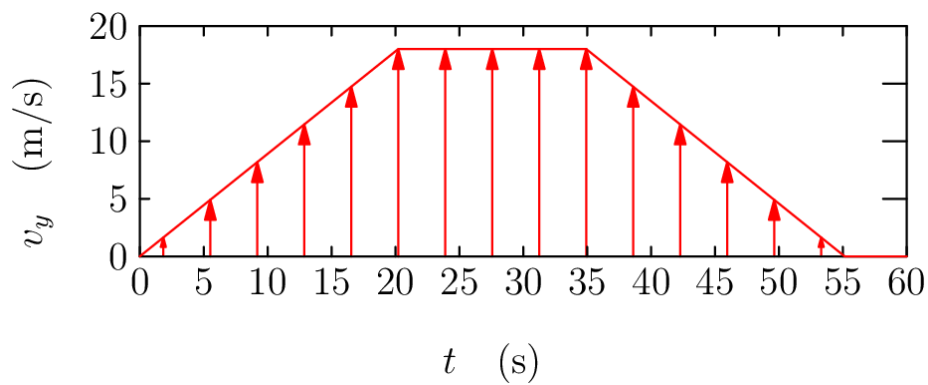
aninconle 2026-09-18 19:49:26



aninconle 2026-09-18 19:49:30

To make a graph, we connect the arrow tops.

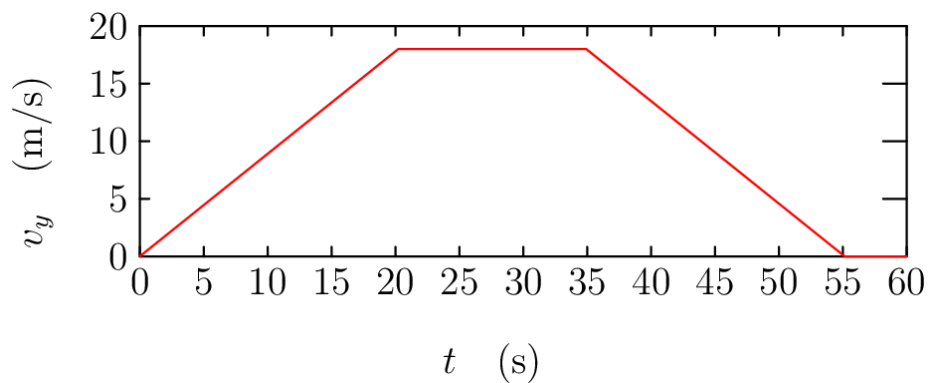
aninconle 2026-09-18 19:49:31



aninconle 2026-09-18 19:49:33

Then we get rid of the arrows:

aninconle 2026-09-18 19:49:34



aninconle 2026-09-18 19:49:39

According to this graph, at what times is the elevator moving in the upward direction?

agoel2010 2026-09-18 19:52:23

0 to 55 s

Thorongil 2026-09-18 19:52:23

from $t = 0$ to $t = 55$

Diamond_Zombie67 2026-09-18 19:52:23

0-55

Zip-a-Dee-Doo-Dah 2026-09-18 19:52:23

*from 0 to 55.

guacataj 2026-09-18 19:52:23

0-55 s

Bunnicorn 2026-09-18 19:52:23

0-55

flybirdeee_ 2026-09-18 19:52:23

0-55 s

vmene 2026-09-18 19:52:23

from 0 to 55

lullipvella 2026-09-18 19:52:23

(0,55)

M.Zhang555 2026-09-18 19:52:23

0 to 55

nickypicky39084 2026-09-18 19:52:23

0s-55s

Ash_Ketchup13 2026-09-18 19:52:23

0s-55s

starkmh002 2026-09-18 19:52:27

oh actually 0-55

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It's all the times from 0 s to 55 s. Whenever the velocity is positive, the elevator is going up.

aninconle 2026-09-18 19:52:33

The elevator doesn't go down in the period represented by this plot. What would it look like if the elevator did go down?

Alexmo2022 2026-09-18 19:54:18

negative velocity

anisquash 2026-09-18 19:54:18

The velocity time graph will be below the horizontal time axis

Oceanix 2026-09-18 19:54:18

it would have a negative velocity

syl314 2026-09-18 19:54:18

Negative velocity (it would fall off our graph)

mathisfun2026 2026-09-18 19:54:18

it would be below the axis

AN29 2026-09-18 19:54:18

it would cross the x axis

apollo2000 2026-09-18 19:54:18

The velocity would be negative

Thorongil 2026-09-18 19:54:18

the velocity would be negative, so the red would be below the t -axis

hw_zasf.xAx 2026-09-18 19:54:18

into the negative y axis

lullipvella 2026-09-18 19:54:18

Like a reflection across the horizontal axis

starkmh002 2026-09-18 19:54:18

it would be negative m/s i think

Supertomato1919 2026-09-18 19:54:18

it goes NEGATIVE!!!

M.Zhang555 2026-09-18 19:54:18

the velocity is negative

LCLARK5691 2026-09-18 19:54:18

the velocity would go below 0

Potatomiaomi 2026-09-18 19:54:18

velocity would be negative

VeryDragon2013 2026-09-18 19:54:18

It would have negative values in the y-axis

Canada1334 2026-09-18 19:54:18

The graph would be opposite; negative

Supertomato1919 2026-09-18 19:54:18

FLIPPED BY THE X-AXIS

aninconle 2026-09-18 19:54:20

The velocity would be negative, and the graph would go beneath the horizontal axis.

aninconle 2026-09-18 19:54:22

Acceleration from a Velocity Graph

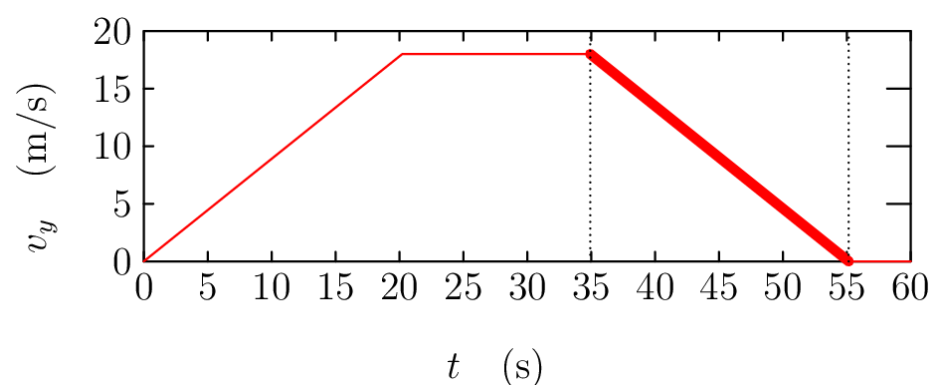
aninconle 2026-09-18 19:54:23

Use a velocity versus time graph to find accelerations.

aninconle 2026-09-18 19:54:25

There is a section from 35 s to 55 s where the graph slopes downward.

aninconle 2026-09-18 19:54:26



aninconle 2026-09-18 19:54:28

What is the elevator doing in this segment?

starkmh002 2026-09-18 19:55:42

slowing down

Zip-a-Dee-Doo-Dah 2026-09-18 19:55:42

slowing down.

flybirdeee_ 2026-09-18 19:55:42

decelerating/slowing down

apollo2000 2026-09-18 19:55:42

The elevator is slowing down.

Oceanix 2026-09-18 19:55:42

slowing down

hw_zasf.xAx 2026-09-18 19:55:42

slowing down

kanishka.chittoor 2026-09-18 19:55:42

slowing down

mocha_x 2026-09-18 19:55:42

slowing down it's acceleration

Nebula233 2026-09-18 19:55:42

slowing down

Grace_174 2026-09-18 19:55:42

slowing down

kabir_sahil 2026-09-18 19:55:42

Slowing down

Brightgiraffe 2026-09-18 19:55:42

slowing down

CAAI 2026-09-18 19:55:42

slowing down

Alexmo2022 2026-09-18 19:55:42

slowing down

Bunnicorn 2026-09-18 19:55:42

slowing down

Potatomiaomi 2026-09-18 19:55:42

slowing down

TatuShark 2026-09-18 19:55:42

slowing down

aninconle 2026-09-18 19:55:55

It's going up, but slowing down. Since the velocity is changing, the elevator has *acceleration*.

aninconle 2026-09-18 19:55:57

What units will we use to measure acceleration?

kanishka.chittoor 2026-09-18 19:57:05

m/s^2

Ash_Ketchup13 2026-09-18 19:57:05

m/s^2

Oceanix 2026-09-18 19:57:05

m/s^2

Potatomiaomi 2026-09-18 19:57:05

$m/(s^2)$

LCLARK5691 2026-09-18 19:57:05

m/s^2

Thorongil 2026-09-18 19:57:05

m/s^2

apollo2000 2026-09-18 19:57:05

meters per seconds²

TatuShark 2026-09-18 19:57:05

m/s^2

Brightgiraffe 2026-09-18 19:57:05

m/s^2

Grace_174 2026-09-18 19:57:05

m/s^2

lullipvella 2026-09-18 19:57:05

$(m/s)/s$

LCLARK5691 2026-09-18 19:57:05

$$\frac{m}{s^2}$$

Seifr 2026-09-18 19:57:05

$$m/s^2$$

loripachiardi 2026-09-18 19:57:05

$$(m/s^2)$$

hw_zasf.xAx 2026-09-18 19:57:05

$$m/s^2$$

aninconle 2026-09-18 19:57:07

The acceleration is

aninconle 2026-09-18 19:57:08

$$a_{y;avg} = \frac{\Delta v_y}{\Delta t}.$$

aninconle 2026-09-18 19:57:09

Our units are the units of v_y divided by the units of t .

$$\frac{m/s}{s} = \frac{m}{s^2}.$$

aninconle 2026-09-18 19:57:12

The Δ symbols mean "change in," which doesn't affect the units. These units are called "meters per second squared."

aninconle 2026-09-18 19:57:16

What was the average acceleration of the elevator between 35 s and 55 s?

thekitsune 2026-09-18 19:59:13

$$-0.9m/s^2$$

Grace_174 2026-09-18 19:59:13

$$-0.9 m/s^2$$

Ash_Ketchup13 2026-09-18 19:59:13

$$-0.9 m/s^2$$

aninconle 2026-09-18 19:59:16

The velocity went from 18 m/s down to 0 m/s, so

$$\begin{aligned}\Delta v_y &= 0 \text{ m/s} - 18 \text{ m/s} \\ &= -18 \text{ m/s}.\end{aligned}$$

aninconle 2026-09-18 19:59:18

The time was

$$\begin{aligned}\Delta t &= 55 \text{ s} - 35 \text{ s} \\ &= 20 \text{ s}.\end{aligned}$$

aninconle 2026-09-18 19:59:19

That makes the average acceleration

$$\begin{aligned}
 a_{y;\text{avg}} &= \frac{\Delta v_y}{\Delta t} \\
 &= \frac{-18 \text{ m/s}}{20 \text{ s}} \\
 &= -0.9 \text{ m/s}^2.
 \end{aligned}$$

aninconle 2026-09-18 19:59:23

Let's interpret this acceleration in terms of the plot. Does Δv_y show up as some particular distance on the plot?

Ash_Ketchup13 2026-09-18 20:01:29

It shows up as distance between 35s and 55s

Seifr 2026-09-18 20:01:29

The y axis is the velocity.

Alexmo2022 2026-09-18 20:01:29

the "rise" in rise/run=slope

AN29 2026-09-18 20:01:29

the distance between the y value of the first and last points

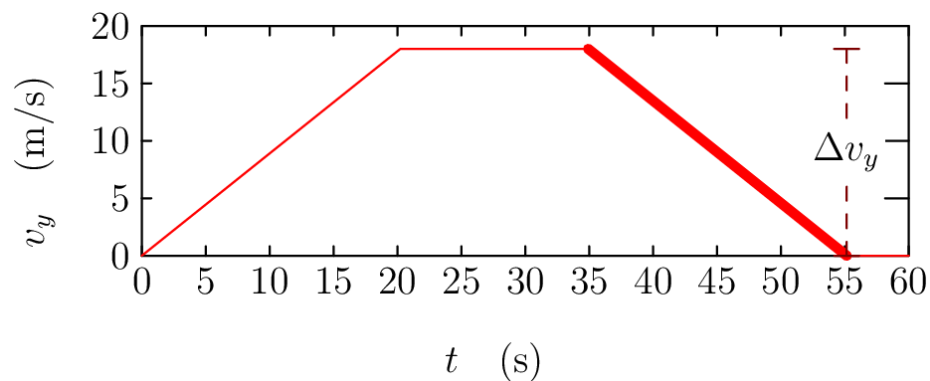
thekitsune 2026-09-18 20:01:29

vertical

aninconle 2026-09-18 20:01:30

It's the vertical distance of the thick segment: the rise (which happens to be negative).

aninconle 2026-09-18 20:01:32



aninconle 2026-09-18 20:01:33

And what's the interpretation of Δt on the graph?

Alexmo2022 2026-09-18 20:02:29

the "run"

Thorongil 2026-09-18 20:02:29

the run

thekitsune 2026-09-18 20:02:29

run - horizontal

hw_zasf.xAx 2026-09-18 20:02:29

the run in slope

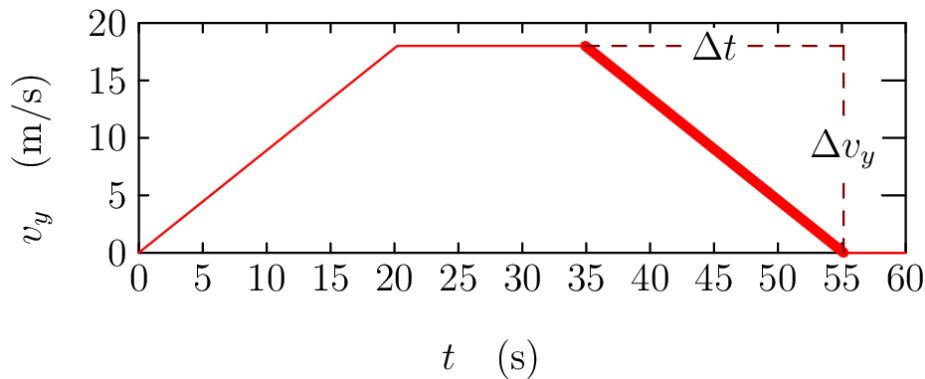
agoel2010 2026-09-18 20:02:29

run

aninconle 2026-09-18 20:02:31

It's the run.

aninconle 2026-09-18 20:02:32



aninconle 2026-09-18 20:02:37

So what is $a_{y;avg}$, interpreted as a feature of the plot?

Seifr 2026-09-18 20:03:38

The slope of the graph.

Potatomiaomi 2026-09-18 20:03:38

slope

Canada1334 2026-09-18 20:03:38

the slope

Alexmo2022 2026-09-18 20:03:38

slope

apollo2000 2026-09-18 20:03:38

The "slope" of the velocity

anisquash 2026-09-18 20:03:38

rise / run so the slope

Grace_174 2026-09-18 20:03:38

rise over run

lullipvella 2026-09-18 20:03:38

The slope

Brightgiraffe 2026-09-18 20:03:38

rise/run or the slope

Zip-a-Dee-Doo-Dah 2026-09-18 20:03:38

rise over run = the slope

Oceanix 2026-09-18 20:03:38

slope

Diamond_Zombie67 2026-09-18 20:03:38

the slope of the line

flybirdeee_ 2026-09-18 20:03:38

the slope

VeryDragon2013 2026-09-18 20:03:38

The slope of the line

TatuShark 2026-09-18 20:03:38

the slope of a segment

aahelic13 2026-09-18 20:03:38

the slope

aninconle 2026-09-18 20:03:42

$a_{y;avg} = \frac{\Delta v_y}{\Delta t}$, or $\frac{\text{rise}}{\text{run}}$ for that thick segment. That's the *slope* of the segment.

aninconle 2026-09-18 20:03:44

What's the acceleration of the elevator between 20 s and 35 s?

lullipvella 2026-09-18 20:04:49

0

anisquash 2026-09-18 20:04:49

0 m/s^2 no acceleration

lullipvella 2026-09-18 20:04:49

0m/s^2

mathisfun2026 2026-09-18 20:04:49

0m/s^2

Oceanix 2026-09-18 20:04:49

0 m/s^2

apollo2000 2026-09-18 20:04:49

0 m/s^2

pandafrog 2026-09-18 20:04:49

0 m/s^2

Alexmo2022 2026-09-18 20:04:49

0m/s^2

Potatomiaomi 2026-09-18 20:04:49

0

LCLARK5691 2026-09-18 20:04:49

0

Seifr 2026-09-18 20:04:49

$0m/s^2$

VeryDragon2013 2026-09-18 20:04:49

0

hw_zasf.xAx 2026-09-18 20:04:49

zero

nickypicky39084 2026-09-18 20:04:49

0m/s^2

Ash_Ketchup13 2026-09-18 20:04:49

0

Diamond_Zombie67 2026-09-18 20:04:49

0 m/s^2

Nebula233 2026-09-18 20:04:49

0 m/s^2

TatuShark 2026-09-18 20:04:49

0 m/s^2

guacataj 2026-09-18 20:04:49

0 m/s²

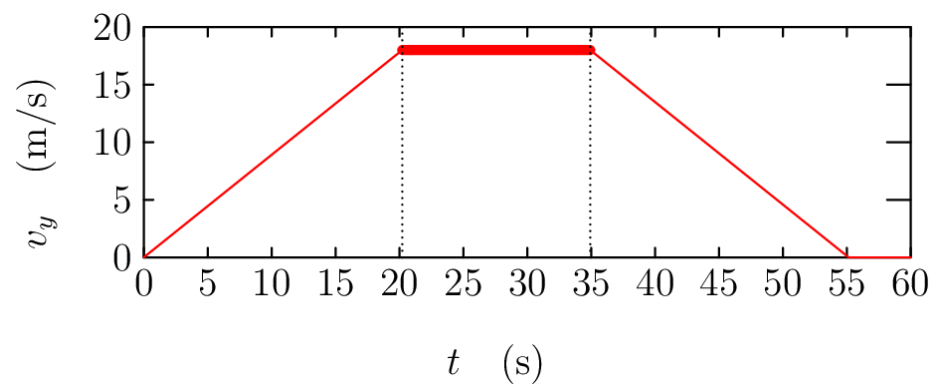
Thorongil 2026-09-18 20:04:49

0

aninconle 2026-09-18 20:04:52

It's 0 m/s²; the graph is flat there.

aninconle 2026-09-18 20:04:53



aninconle 2026-09-18 20:04:57

Answer a series of short questions on acceleration, velocity, and speed.

aninconle 2026-09-18 20:04:59

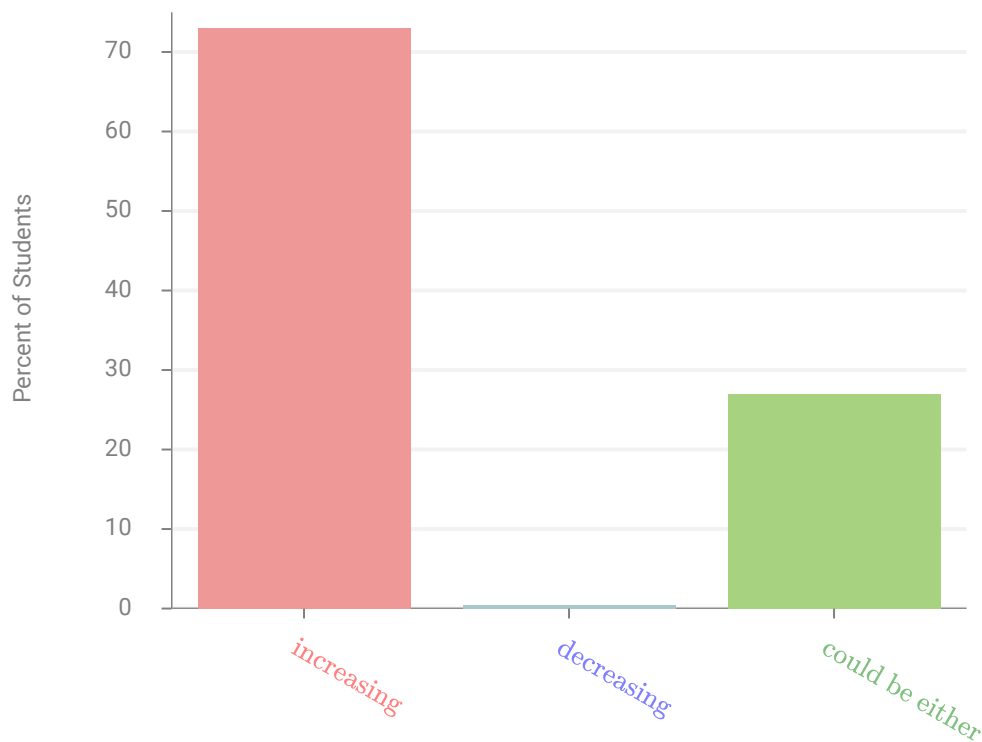
An object's acceleration is positive. Is its speed increasing or decreasing?

increasing

decreasing

could be either

aninconle 2026-09-18 20:05:34



aninconle 2026-09-18 20:05:36

Hmmm

aninconle 2026-09-18 20:05:39

If the acceleration is positive, what does that tell us about the velocity versus time graph?

Exponent11 2026-09-18 20:07:03

increasing?

kanishka.chittoor 2026-09-18 20:07:03

increasing

agoel2010 2026-09-18 20:07:03

slope is postive

TatuShark 2026-09-18 20:07:03

The slope is positive

starkmh002 2026-09-18 20:07:03

velocity is increasing or going up hte y axis

syl314 2026-09-18 20:07:03

It's sloping up

Seifr 2026-09-18 20:07:03

The velocity is increasing.

Alexmo2022 2026-09-18 20:07:03

slope is positive on v vs t graph

Thorongil 2026-09-18 20:07:03

the slope is positive at that point. Therefore, it is increasing at that point.

M.Zhang555 2026-09-18 20:07:03

slope is positive

Bunnicorn 2026-09-18 20:07:03

the slope is positive

LCLARK5691 2026-09-18 20:07:03

the slope must always be positive

Marshall_Huang 2026-09-18 20:07:09

The graph slopes up

aninconle 2026-09-18 20:07:10

The velocity versus time graph slopes upward.

aninconle 2026-09-18 20:07:13

The question asked about *speed*, not *velocity*. Speed is how far the graph is from zero.

aninconle 2026-09-18 20:07:16

Can the slope be positive while a graph is getting closer to zero? If so, how?

agoel2010 2026-09-18 20:08:45

if its going in the oppsite direction or if velocity is negative

AN29 2026-09-18 20:08:45

yes if the velocity is negative

kabir_sahil 2026-09-18 20:08:45

When it starts of negative

hw_zasf.xAx 2026-09-18 20:08:45

if it starts in the negative y axis

Thorongil 2026-09-18 20:08:45

yes, if the y -value is negative

Seifr 2026-09-18 20:08:45

If the velocity is already negative.

Alexmo2022 2026-09-18 20:08:45

Yes, if negative velocity slows and gets closer to zero

apollo2000 2026-09-18 20:08:45

yes, if the velocity was negative and is becoming positive.

starkmh002 2026-09-18 20:08:45

it has to be going from negative acceleration

mathisfun2026 2026-09-18 20:08:45

yes it can go from negative to less negative

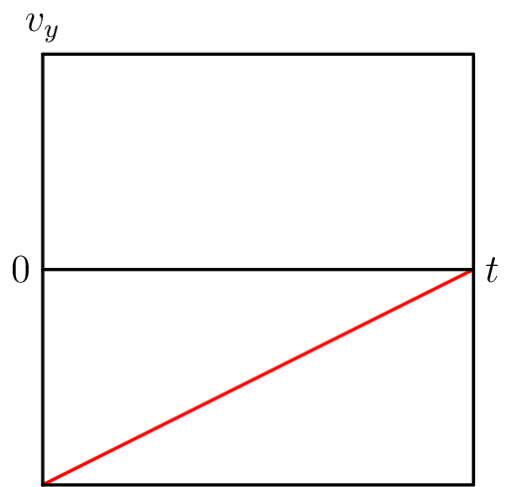
anisquash 2026-09-18 20:08:45

The graph starts off with a really high in magnitude negative velocity, (negative velocity only means speeding in opposite direction) and then slowly reaches the horizontal time axis because the graph is below the time axis. So object starts high in speed and slow down to zero and to a rest.

aninconle 2026-09-18 20:08:49

Yes. This happens when the velocity is negative.

aninconle 2026-09-18 20:08:50



aninconle 2026-09-18 20:09:13

Let's try that again:

aninconle 2026-09-18 20:09:14

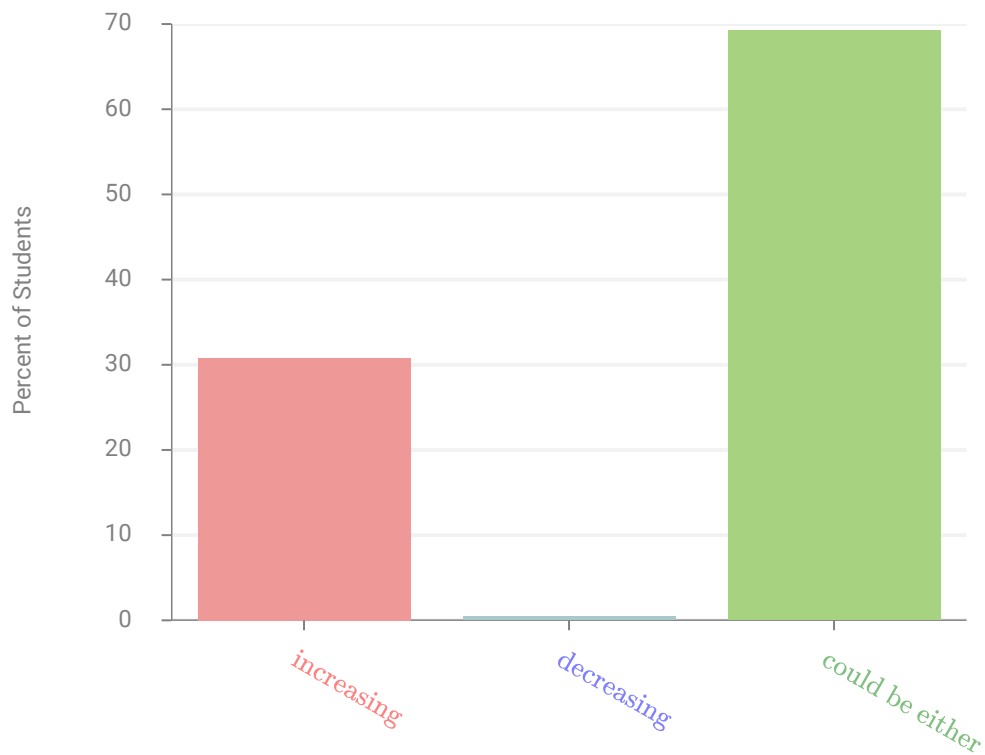
An object's acceleration is positive. Is its speed increasing or decreasing?

increasing

decreasing

could be either

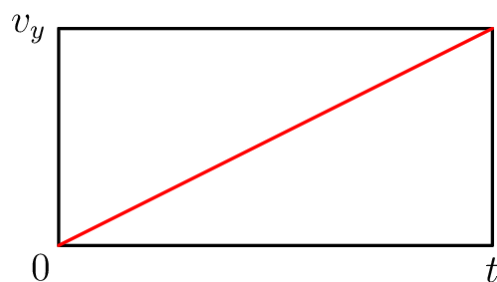
aninconle 2026-09-18 20:09:56



aninconle 2026-09-18 20:10:01

In our previous plot, the acceleration was positive, but the speed was decreasing.

aninconle 2026-09-18 20:10:02



aninconle 2026-09-18 20:10:04

In this plot, the acceleration is positive and speed is increasing.

aninconle 2026-09-18 20:10:08

Positive acceleration means the *velocity* is increasing, but speed can increase or decrease with positive acceleration.

aninconle 2026-09-18 20:10:15

An elevator has *negative* acceleration and *increasing* speed. What's the elevator doing?

aninconle 2026-09-18 20:10:32

What's your answer? Try to be a bit detailed; don't just say whether it's going up or down.

M.Zhang555 2026-09-18 20:12:36

but its accelerating down as well

anisquash 2026-09-18 20:12:36

The elevator is going down.

Ash_Ketchup13 2026-09-18 20:12:36

The elevator is going down faster

aahelic13 2026-09-18 20:12:36

moving downward while speeding up

Oceanix 2026-09-18 20:12:36

the elevator going down at a faster and faster pace

Thorongil 2026-09-18 20:12:36

going downwards, but speeding up as it goes downwards

flybirdeee_ 2026-09-18 20:12:36

the elevator is moving downward with increasing speed

Grace_174 2026-09-18 20:12:36

The elevator is going up but slowing down

kanishka.chittoor 2026-09-18 20:12:36

down because if the acceleration is negative, that means its speed is increasing then it's going down

kabir_sahil 2026-09-18 20:12:36

And it is gradually increasing speed while going down

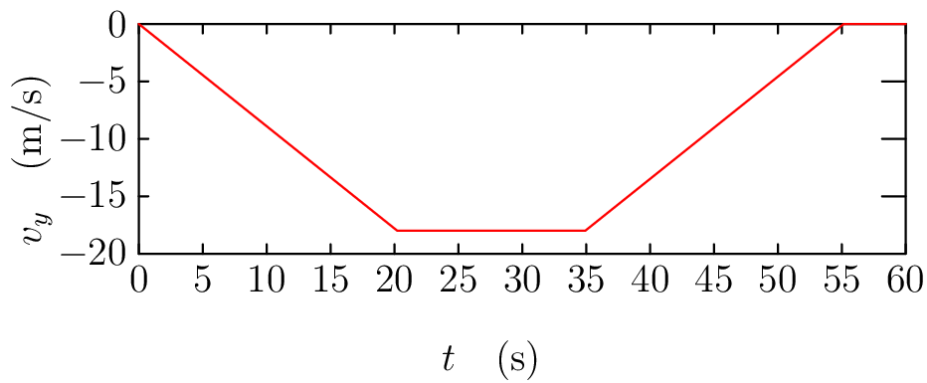
aninconle 2026-09-18 20:12:40

It's probably the start of a downward trip. Let's make a velocity versus time graph for a downward trip. How can we do that, starting from our plot for an upward trip?

aninconle 2026-09-18 20:12:43

We multiply all the velocities by -1 .

aninconle 2026-09-18 20:12:46



aninconle 2026-09-18 20:12:50

Between 0 s and 20 s, the elevator's speed is increasing, but the acceleration is negative because the plot slopes down.

aninconle 2026-09-18 20:12:51

Calculating with Acceleration

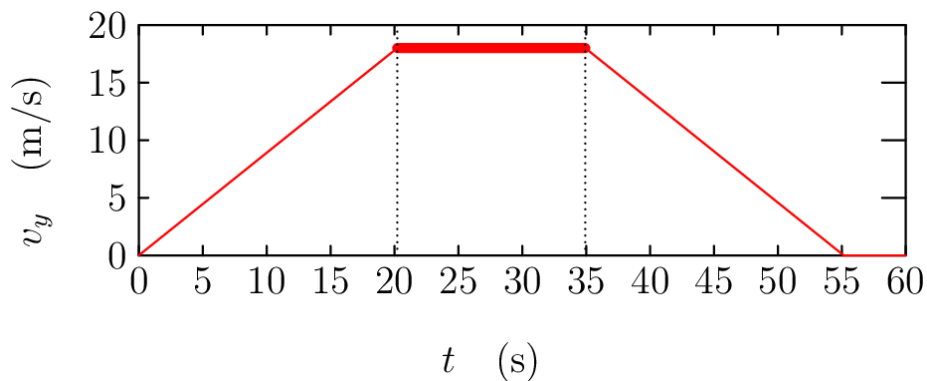
aninconle 2026-09-18 20:12:52

Imagine you could get rid of the constant speed section in the Shanghai Tower elevator. How long would the ride be then?

aninconle 2026-09-18 20:13:03

Our plot has a flat section.

aninconle 2026-09-18 20:13:04



aninconle 2026-09-18 20:13:06

This question is asking us to imagine that the elevator keeps accelerating instead of flatlining. If so, what would the velocity versus time curve look like?

Nebula233 2026-09-18 20:15:00

like a triangle

syl314 2026-09-18 20:15:00

A triangle

VeryDragon2013 2026-09-18 20:15:00

just like a flipped absolute value

leopards 2026-09-18 20:15:00

upside down v

Zip-a-Dee-Doo-Dah 2026-09-18 20:15:00

It would be a triangle instead.

nickypicky39084 2026-09-18 20:15:00

upside down V

hhatchell 2026-09-18 20:15:00

It would make a triangle?

apollo2000 2026-09-18 20:15:00

It would look like a triangle

hw_zasf.xAx 2026-09-18 20:15:00

like the top part of a triangle ^

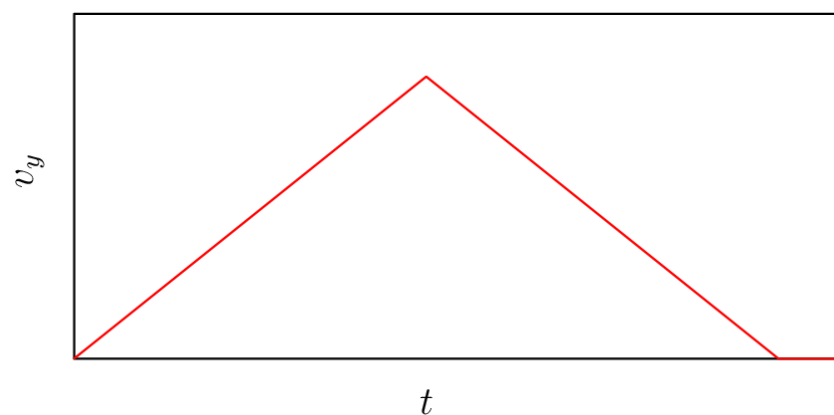
AN29 2026-09-18 20:15:00

it would be an upside down v

aninconle 2026-09-18 20:15:02

It would be an upside-down V shape.

aninconle 2026-09-18 20:15:03



aninconle 2026-09-18 20:15:04

There are no axis labels because we don't know the max speed or total trip time. What can we introduce to represent those unknowns as we solve this problem?

Nebula233 2026-09-18 20:16:22

variables?

apollo2000 2026-09-18 20:16:22

variables

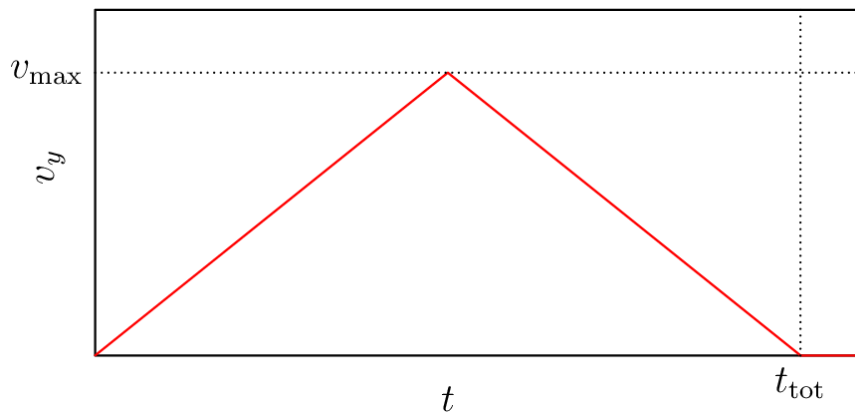
AN29 2026-09-18 20:16:22

variables?

aninconle 2026-09-18 20:16:23

We can use variables. We'll call the maximum speed $v_{\max}$ and the total time of the ride t_{tot} .

aninconle 2026-09-18 20:16:25



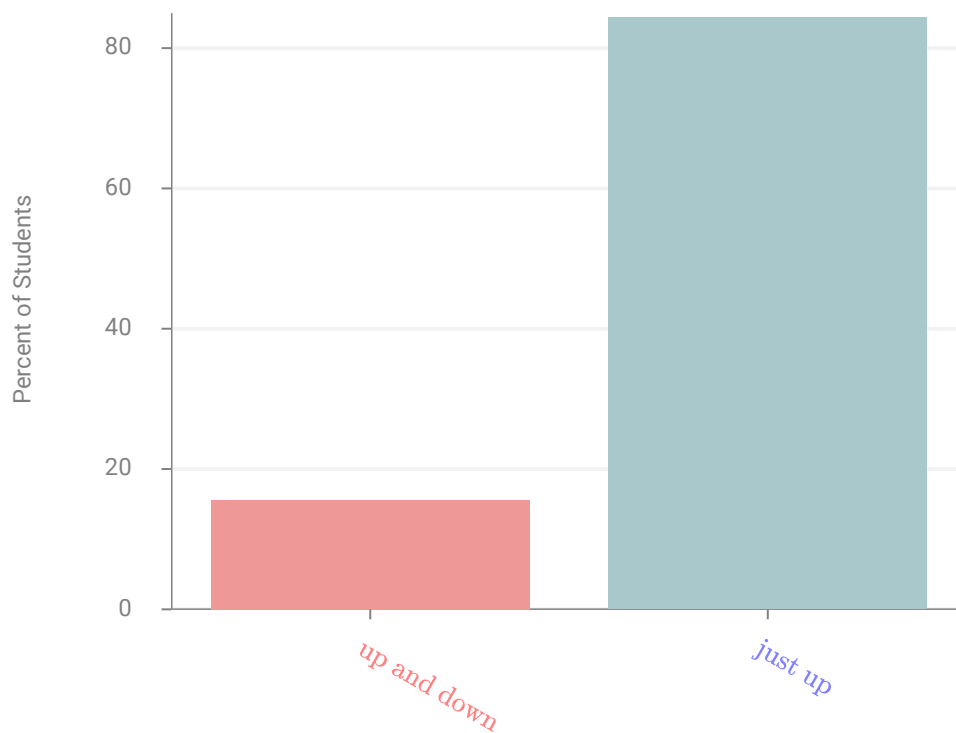
aninconle 2026-09-18 20:16:27

Remind me: is this graph for a trip up and down the building, or just up?

up and down

just up

aninconle 2026-09-18 20:17:17



aninconle 2026-09-18 20:17:18

It's just up. The velocity curve goes up and down, but it's positive the whole time—always going up.

aninconle 2026-09-18 20:17:19

The magnitude of acceleration can be a_0 . Then $a_y = a_0$ for the first half and $a_y = -a_0$ for the second half.

aninconle 2026-09-18 20:17:20

Although we know $a_0 = 0.9 \text{ m/s}^2$, it will make sense to work with a variable until the end of our calculations, since $v_{\max}$ and t_{tot} are also variables.

aninconle 2026-09-18 20:17:21

We have

$v_{\max}$: maximum speed of elevator

t_{tot} : total duration of trip

a_0 : magnitude of acceleration.

aninconle 2026-09-18 20:17:23

Are there any other quantities we'll need in order to solve this problem?

flybirddee_ 2026-09-18 20:18:49

distance?

anisquash 2026-09-18 20:18:49

The displacement

Zip-a-Dee-Doo-Dah 2026-09-18 20:18:49

distance..?

thekitsune 2026-09-18 20:18:49

displacement

aninconle 2026-09-18 20:18:56

Which distance, specifically?

aahelic13 2026-09-18 20:20:37

distance from the first floor to last floor

lullipvella 2026-09-18 20:20:37

the height of the tower

Thorongil 2026-09-18 20:20:37

the height of the Shanghai tower

kanishka.chittoor 2026-09-18 20:20:37

height of tower

LCLARK5691 2026-09-18 20:20:37

the distance from the bottom to the top of the elevator

leopards 2026-09-18 20:20:37

how high the Shanghai tower elevator needs to travel to reach the top

Nebula233 2026-09-18 20:20:37

from bottom of building to top

Zip-a-Dee-Doo-Dah 2026-09-18 20:20:37

The distance it takes to go from the bottom to the top of the elevator path.

LCLARK5691 2026-09-18 20:20:37

the distance from the bottom to the top of the building

Seifr 2026-09-18 20:20:37

Total distance travelled by the elevator.

Oceanix 2026-09-18 20:20:37

distance from bottom of the elevator to the top

AN29 2026-09-18 20:20:37

the height of the tower

starkmh002 2026-09-18 20:20:37

the total height of the building

aninconle 2026-09-18 20:20:39

The total height of the skyscraper also matters. Let's call that H .

aninconle 2026-09-18 20:20:40

variable	meaning	value
$v_{\max}$	maximum speed reached	?
t_{tot}	total duration of trip	?
a_0	magnitude of acceleration	0.9 m/s ²
H	total height of trip	630 m

aninconle 2026-09-18 20:20:44

We're looking for t_{tot} .

aninconle 2026-09-18 20:20:55

If you're looking for a time to travel a certain distance, what physical quantity do you normally use?

aninconle 2026-09-18 20:22:30

We need the average velocity over the entire trip. Then we can use

$$\Delta y = v_{y;\text{avg}} \Delta t.$$

aninconle 2026-09-18 20:22:33

In this problem, what do we use instead of Δy ?

Thorongil 2026-09-18 20:23:47

H

AN29 2026-09-18 20:23:47

H

kanishka.chittoor 2026-09-18 20:23:47

H

lullipvella 2026-09-18 20:23:47

H

TatuShark 2026-09-18 20:23:47

H?

VeryDragon2013 2026-09-18 20:23:47

H

M.Zhang555 2026-09-18 20:23:47

use H

leopards 2026-09-18 20:23:47

H

aninconle 2026-09-18 20:23:48

We use H , the height. We also use t_{tot} in place of Δt . Also, since the velocity is always positive, speed and velocity are the same here.

$$H = v_{\text{avg}} t_{\text{tot}}.$$

aninconle 2026-09-18 20:23:53

What's the average speed in terms of a variable on our stickied list?

leopards 2026-09-18 20:25:05

$v_{\max} / 2$

Ash_Ketchup13 2026-09-18 20:25:05

$v_{\max}/2$

kabir_sahil 2026-09-18 20:25:05

$v_{\max}/2$

M.Zhang555 2026-09-18 20:25:05

$1/2v_{\max}$

aninconle 2026-09-18 20:25:10

The speed starts at zero and rises steadily up to $v_{\max}$. The average speed for the first half is $\frac{1}{2}v_{\max}$. The same works for the second half, so the average speed is

aninconle 2026-09-18 20:25:12

$$v_{\text{avg}} = \frac{1}{2}v_{\max}.$$

aninconle 2026-09-18 20:25:14

That means

aninconle 2026-09-18 20:25:15

$$H = \left(\frac{1}{2}v_{\max} \right) t_{\text{tot}}.$$

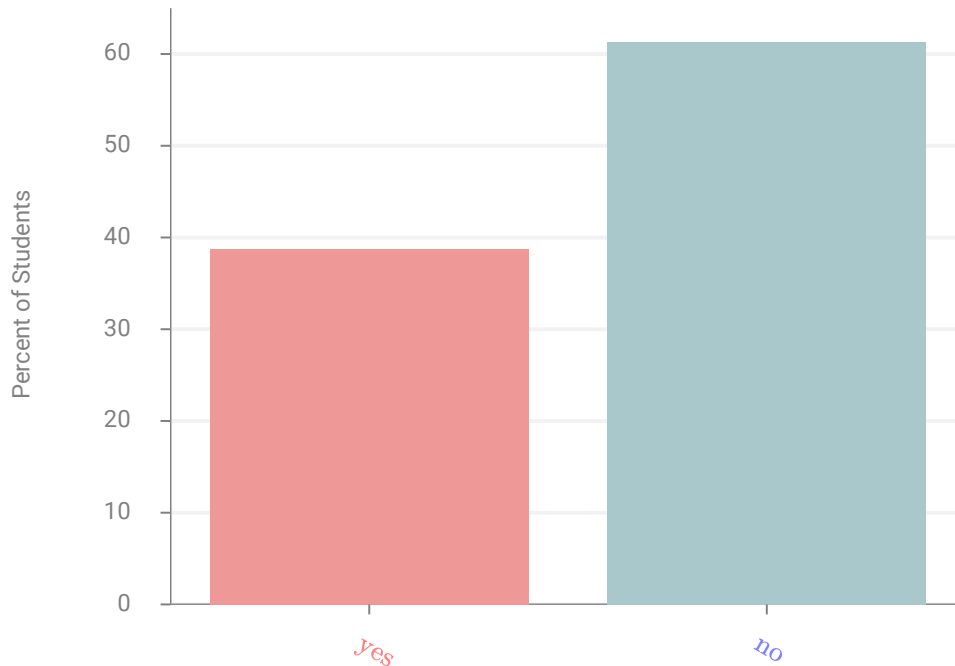
aninconle 2026-09-18 20:25:21

Are we ready to plug in numbers and find t_{tot} ?

yes

no

aninconle 2026-09-18 20:26:24



aninconle 2026-09-18 20:26:32

We don't know $v_{\max}$ yet.

aninconle 2026-09-18 20:26:33

What two variables can we likely combine to get $v_{\max}$?

flybirdeee_ 2026-09-18 20:27:14

acceleration and time

Thorongil 2026-09-18 20:27:14

a_0 and t_{tot} ?

VeryDragon2013 2026-09-18 20:27:14

t_{tot} , a_0

aninconle 2026-09-18 20:27:15

Since acceleration tells us how quickly the velocity is changing over time, we should be able to combine a_0 and t_{tot} to get $v_{\max}$.

aninconle 2026-09-18 20:27:17

Do you see what $v_{\max}$ is in terms of a_0 and t_{tot} ?

aninconle 2026-09-18 20:27:24

(Hint: a_0 is the slope of the first part of the velocity versus time curve.)

VeryDragon2013 2026-09-18 20:28:27

$$\frac{a_0 t_{\text{tot}}}{2}$$

leopards 2026-09-18 20:28:27

$1/2 \cdot a_0 \cdot t_{\text{tot}}$?

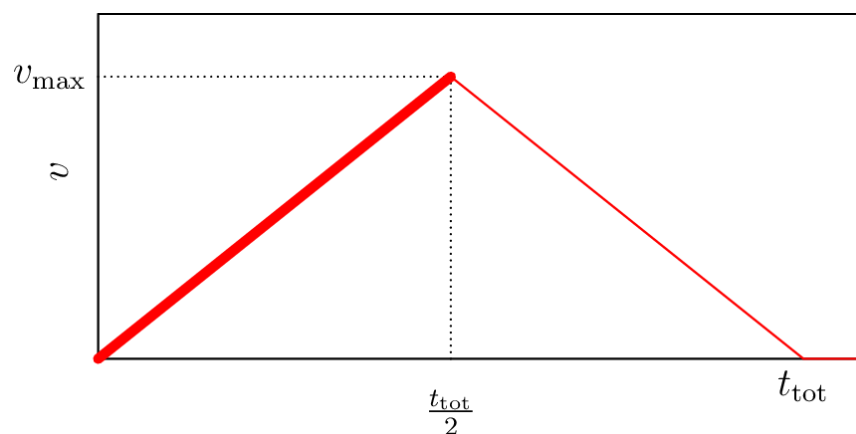
syl314 2026-09-18 20:28:27

$$a_0 \cdot \frac{t_{\text{tot}}}{2}$$

aninconle 2026-09-18 20:28:28

The first part of the velocity versus time curve has a slope of a_0 , and it stretches over a duration $\frac{t_{\text{tot}}}{2}$.

aninconle 2026-09-18 20:28:29



aninconle 2026-09-18 20:28:30

Since the slope is a_0 , we can say

aninconle 2026-09-18 20:28:31

$$a_0 = \frac{\Delta v}{\Delta t}$$

$$= \frac{v_{\max}}{t_{\text{tot}}/2}.$$

aninconle 2026-09-18 20:28:33

When we solve this for $v_{\max}$, we get

aninconle 2026-09-18 20:28:34

$$v_{\max} = a_0 \cdot \frac{t_{\text{tot}}}{2}.$$

aninconle 2026-09-18 20:28:42

Our goal is to find t_{tot} in terms of H and a_0 .

aninconle 2026-09-18 20:28:44

We can take the equation for $v_{\max}$ and plug that into the equation for H . Then we can solve for t_{tot} .

aninconle 2026-09-18 20:28:45

What do you get for t_{tot} ?

$$t_{\text{tot}} = \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

Thorongil 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

M.Zhang555 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

LCLARK5691 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

Canada1334 2026-09-18 20:31:31

$$t_{\text{tot}} = a_0 \left(\frac{H}{2} \right)^{\frac{1}{2}}$$

AN29 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{a_0}{H} \right)^{\frac{1}{2}}$$

mathisfun2026 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

Bunnicorn 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

kabir_sahil 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

Brightgiraffe 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

apollo2000 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

VeryDragon2013 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

leopards 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

anisquash 2026-09-18 20:31:31

$$t_{\text{tot}} = 2 \left(\frac{H}{a_0} \right)^{\frac{1}{2}}$$

aninconle 2026-09-18 20:31:34

After substituting $v_{\text{max}} = a_0 \frac{t_{\text{tot}}}{2}$ into $H = \left(\frac{1}{2} v_{\text{max}} \right) t_{\text{tot}}$, we get

$$\begin{aligned} H &= \left(\frac{1}{2} \left(a_0 \frac{t_{\text{tot}}}{2} \right) \right) t_{\text{tot}} \\ &= \frac{1}{4} a_0 t_{\text{tot}}^2. \end{aligned}$$

aninconle 2026-09-18 20:31:35

Then we solve for t_{tot} to get

aninconle 2026-09-18 20:31:36

$$\begin{aligned} t_{\text{tot}} &= \sqrt{\frac{4H}{a_0}} \\ &= 2 \sqrt{\frac{H}{a_0}}. \end{aligned}$$

aninconle 2026-09-18 20:31:43

t_{tot} goes down as we increase a_0 . That makes sense—the greater the acceleration, the less time it will take to get to the top of the building.

aninconle 2026-09-18 20:31:47

For the Shanghai tower, $H = 630$ m and $a_0 = 0.9$ m/s². How long would it take our hypothetical elevator to get to the top?

Thorongil 2026-09-18 20:32:52

about 52.9 s

Nebula233 2026-09-18 20:32:52

~52.92 seconds

apollo2000 2026-09-18 20:32:52

52.92 seconds

leopards 2026-09-18 20:32:52

53 s

VeryDragon2013 2026-09-18 20:32:52

53 seconds?

Ash_Ketchup13 2026-09-18 20:32:52

53 seconds

aninconle 2026-09-18 20:32:54

The time is

$$t_{\text{tot}} = \sqrt{\frac{4 \cdot 630 \text{ m}}{0.9 \text{ m/s}^2}}$$

$$= 52.9 \text{ s.}$$

aninconle 2026-09-18 20:33:00

Before our modification, the elevator ride was 55 s, so how much time have we saved?

leopards 2026-09-18 20:33:48

2s

kabir_sahil 2026-09-18 20:33:48

2.1 secs

kanishka.chittoor 2026-09-18 20:33:48

2.1s

AN29 2026-09-18 20:33:48

around 2s

syl314 2026-09-18 20:33:48

Like 2 seconds

apollo2000 2026-09-18 20:33:48

~2.08 seconds

Zip-a-Dee-Doo-Dah 2026-09-18 20:33:48

about 2 seconds 😊

Bunnicorn 2026-09-18 20:33:48

2.1s

flybirdeee_ 2026-09-18 20:33:48

2.1s

starkmh002 2026-09-18 20:33:48

2.1 seconds

TatuShark 2026-09-18 20:33:48

2.1 s

Ash_Ketchup13 2026-09-18 20:33:48

2.1 seconds

thekitsune 2026-09-18 20:33:48

2.1s

Canada1334 2026-09-18 20:33:48

2.1 seconds

lullipvella 2026-09-18 20:33:48

2.1 s

Marshall_Huang 2026-09-18 20:33:48

2 seconds, not a lot

veryhungrypanda 2026-09-18 20:33:48

2.1 s

hw_zasf.xAx 2026-09-18 20:33:48

2.1seconds

Seifr 2026-09-18 20:33:48

2.1 seconds.

VeryDragon2013 2026-09-18 20:33:48

2.1 seconds! Wow! So much time!

aninconle 2026-09-18 20:33:50

Hardly any—just 2.1 s! The constant velocity portion of the real elevator ride is fairly short. Adding some extra acceleration during that section doesn't accomplish much!

aninconle 2026-09-18 20:33:52

Suppose you wanted to make a constant-acceleration trip up a super skyscraper nine times as tall as the Shanghai Tower. How long would that take, assuming the same a_0 ?

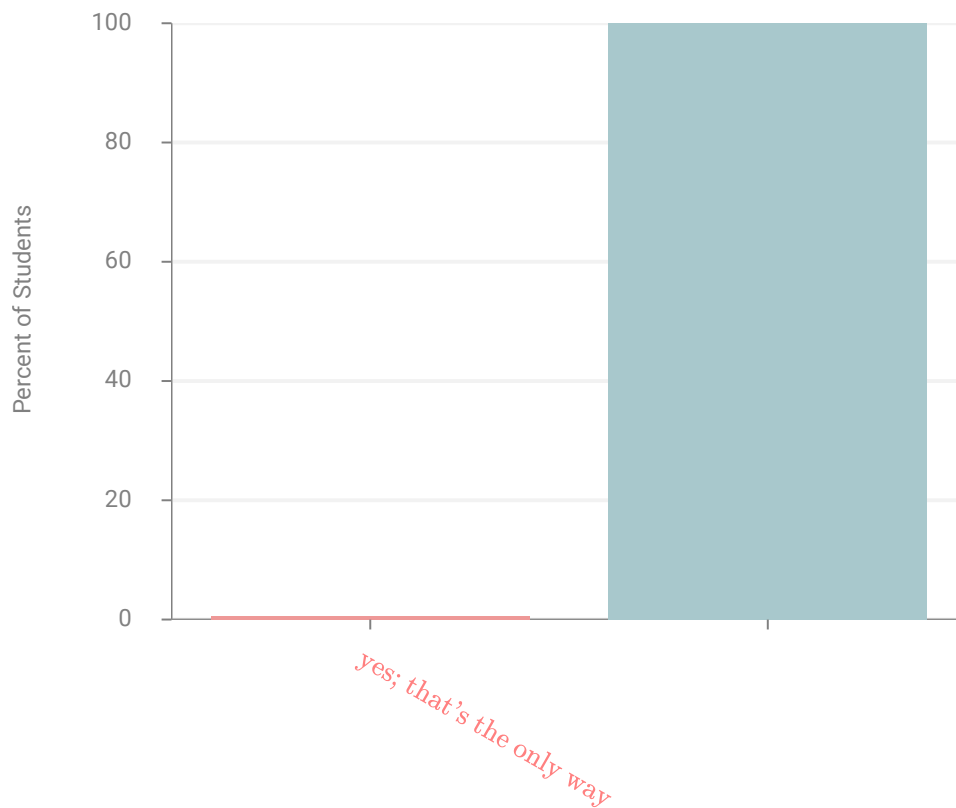
aninconle 2026-09-18 20:33:54

Do we need to plug all new numbers into $t_{\text{tot}} = 2\sqrt{\frac{H}{a_0}}$ and evaluate again?

yes; that's the only way

no; there's a shortcut using our previous calculation

aninconle 2026-09-18 20:35:03



aninconle 2026-09-18 20:35:04

Since we're multiplying the height by nine, our equation becomes

$$t_{\text{tot;super}} = 2\sqrt{\frac{(9H)}{a_0}}$$

$$= 3 \cdot \left(2\sqrt{\frac{H}{a_0}}\right).$$

aninconle 2026-09-18 20:35:10

We already evaluated

$$t_{\text{tot;normal}} = 2\sqrt{\frac{H}{a_0}}$$

$$= 52.9 \text{ s}.$$

Our equation is

$$t_{\text{tot;super}} = 3 \cdot t_{\text{tot;normal}}.$$

aninconle 2026-09-18 20:35:13

How long does it take to get up this super skyscraper?

lullipvella 2026-09-18 20:36:35

158.7 s

Ash_Ketchup13 2026-09-18 20:36:35

158 seconds

Marshall_Huang 2026-09-18 20:36:35

158.7 s

VeryDragon2013 2026-09-18 20:36:35

158.7 seconds

TatuShark 2026-09-18 20:36:35

158.7 s

mocha_x 2026-09-18 20:36:35

158.7S

apollo2000 2026-09-18 20:36:35

158.7s

syl314 2026-09-18 20:36:35

158.7 secs

anisquash 2026-09-18 20:36:35

158.7 seconds

Canada1334 2026-09-18 20:36:35

158.7 seconds

hw_zasf.xAx 2026-09-18 20:36:35

158.7 seconds

flybirdeee_ 2026-09-18 20:36:35

158.7s

leopards 2026-09-18 20:36:35

159 s

thekitsune 2026-09-18 20:36:35

158.7s

Seifr 2026-09-18 20:36:35

158.7 seconds.

Nebula233 2026-09-18 20:36:35

158.7 sec

Grace_174 2026-09-18 20:36:35

158.7s

aninconle 2026-09-18 20:36:36

It takes $52.9 \text{ s} \cdot 3 = 158.7 \text{ s}$.

aninconle 2026-09-18 20:36:38

Does it make intuitive sense that to get up a skyscraper nine times as tall, it takes three times as long with constant acceleration?

Yes, totally!

Sort of, but I'm not sure.

I don't see any mistake in the equations but I don't get why it happens.

That seems really wrong!

aninconle 2026-09-18 20:37:21

Here's some intuition: Imagine we get in the elevator for the super tall skyscraper without knowing how tall it is. We notice that the elevator ride is three times as long as the one for the normal skyscraper.

aninconle 2026-09-18 20:37:29

Because the accelerations are the same, what would you be able to infer about the max speed in the super skyscraper?

VeryDragon2013 2026-09-18 20:38:36

triple max speed

syl314 2026-09-18 20:38:36

The max speed is faster

kanishka.chittoor 2026-09-18 20:38:36

max speed is 3 times as much

flybirdeee_ 2026-09-18 20:38:36

3 times

aahelic13 2026-09-18 20:38:36

it would be three times as much

thekitsune 2026-09-18 20:38:36

3 x faster

aninconle 2026-09-18 20:38:38

The elevator ride is three times as long with the same acceleration, so the max speed is three times as much.

aninconle 2026-09-18 20:38:44

So in the super skyscraper, the max speed (and the average speed) are three times as high, plus the elevator ride is three times as long.

aninconle 2026-09-18 20:38:47

Going three times as fast for three times as long takes you nine times as high!

aninconle 2026-09-18 20:38:51

We solved this problem with the equation $t_{\text{tot}} = 2\sqrt{\frac{H}{a_0}}$, but it didn't matter that there's a 2 out front or an a_0 in the denominator.

We only looked at the relationship between t_{tot} and H .

aninconle 2026-09-18 20:38:53

When we model just the relationship between two variables and ignore all the others, we are using a *proportionality*. We can write

$$t_{\text{tot}} \propto \sqrt{H}.$$

aninconle 2026-09-18 20:38:56

The total time is proportional to the square root of the height of the building.

aninconle 2026-09-18 20:38:59

Proportionalities are especially useful in problems of *scaling*. These are problems where you've already figured out what happens in one scenario, and now want to make everything bigger or smaller in some new scenario. In this case, everything was the same except that we scaled up the height of the building. We'll continue using proportionalities and scaling throughout the class.

aninconle 2026-09-18 20:39:08

Alright! Let's look at:

aninconle 2026-09-18 20:39:09

Free Fall

aninconle 2026-09-18 20:39:11

Quantify free fall on Earth.

aninconle 2026-09-18 20:39:15

NASA, the US space agency, needs to test what will happen to equipment in outer space, meaning it needs to have environments with no air.

aninconle 2026-09-18 20:39:17

They have a particularly big one at the Space Power Facility in Ohio. It's 37 m tall. NASA is capable of sucking almost all the air out this room.

aninconle 2026-09-18 20:39:20

For a TV show, NASA once dropped a bowling ball and a feather at the same time from the top of this facility.

aninconle 2026-09-18 20:39:22

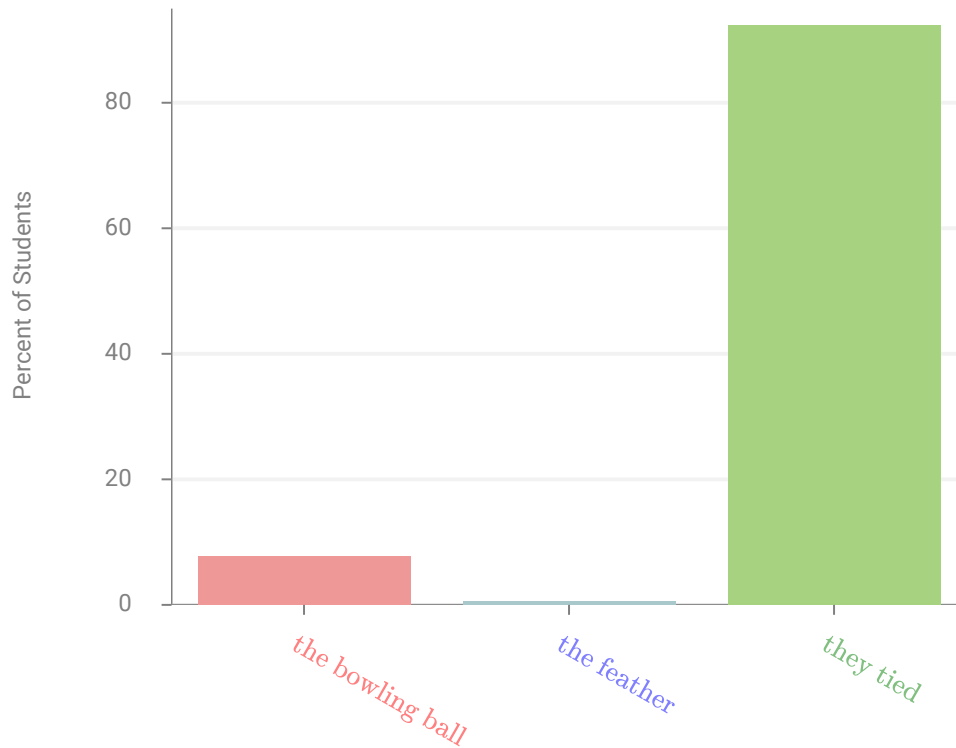
Which item do you think hit the floor first?

the bowling ball

the feather

they tied

aninconle 2026-09-18 20:40:07



aninconle 2026-09-18 20:40:16

Okay, here's the result:

aninconle 2026-09-18 20:40:17



aninconle 2026-09-18 20:40:26

Bowling ball wins, of course!

aninconle 2026-09-18 20:40:35

But wait, is something wrong in this GIF?

starkmh002 2026-09-18 20:42:03

air resistance

Alexmo2022 2026-09-18 20:42:03

there is air resistance

hhatchell 2026-09-18 20:42:03

There's air?

Thorongil 2026-09-18 20:42:03

how is that guy breathing if there's no air?

Ash_Ketchup13 2026-09-18 20:42:03

yes, there is air

VeryDragon2013 2026-09-18 20:42:03

If they sucked out all the air why is there a guy

apollo2000 2026-09-18 20:42:03

Isn't a person in the room?

lullipvella 2026-09-18 20:42:03

The person cannot be there because there is no air!

mathisfun2026 2026-09-18 20:42:03

no the feathers have more air restitance

Sefr 2026-09-18 20:42:03

Their is air in the chamber.

Zip-a-Dee-Doo-Dah 2026-09-18 20:42:03

There appears to be air resistance.

loripachiardi 2026-09-18 20:42:03

that was before they removed the oxygen from the room

Potatomiaomi 2026-09-18 20:42:03

the rooom is not vaccumed

Canada1334 2026-09-18 20:42:03

Also how is the guy breathing if there is no air

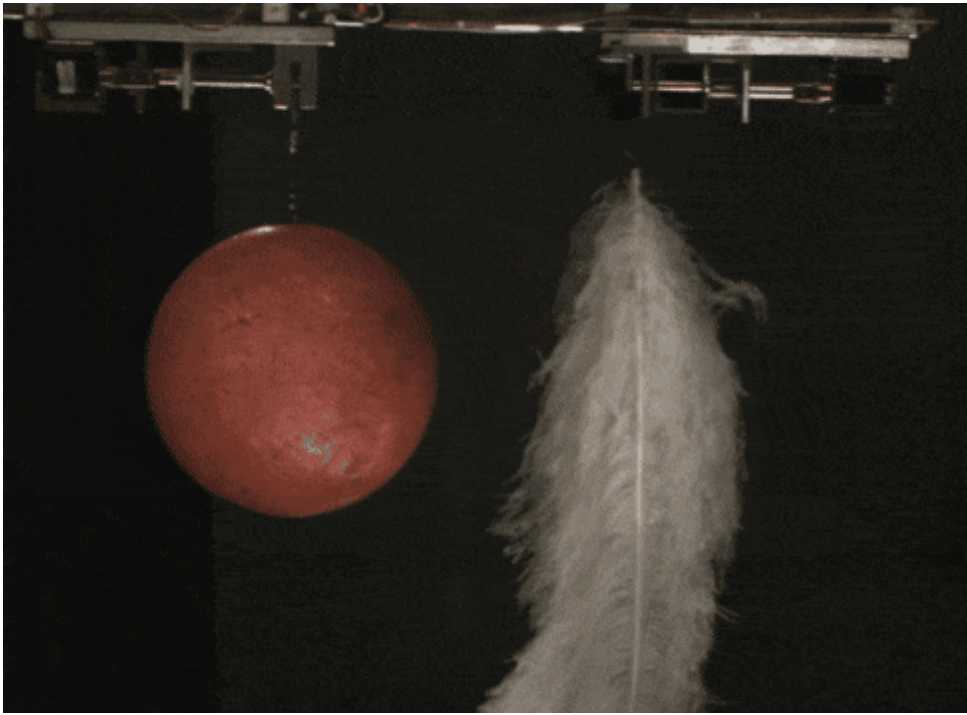
aninconle 2026-09-18 20:42:13

There's a person standing there watching. That person wouldn't be able to breathe if they had sucked the air out. That clip was just the trial run.

aninconle 2026-09-18 20:42:21

Here's what happened after they sucked the air out (shown in slow motion):

aninconle 2026-09-18 20:42:21



aninconle 2026-09-18 20:42:26

A tie!

aninconle 2026-09-18 20:42:28

In the absence of air resistance, any two objects will fall in the same way. This is called *free fall*.

aninconle 2026-09-18 20:42:29

How do you suspect objects in free fall move?

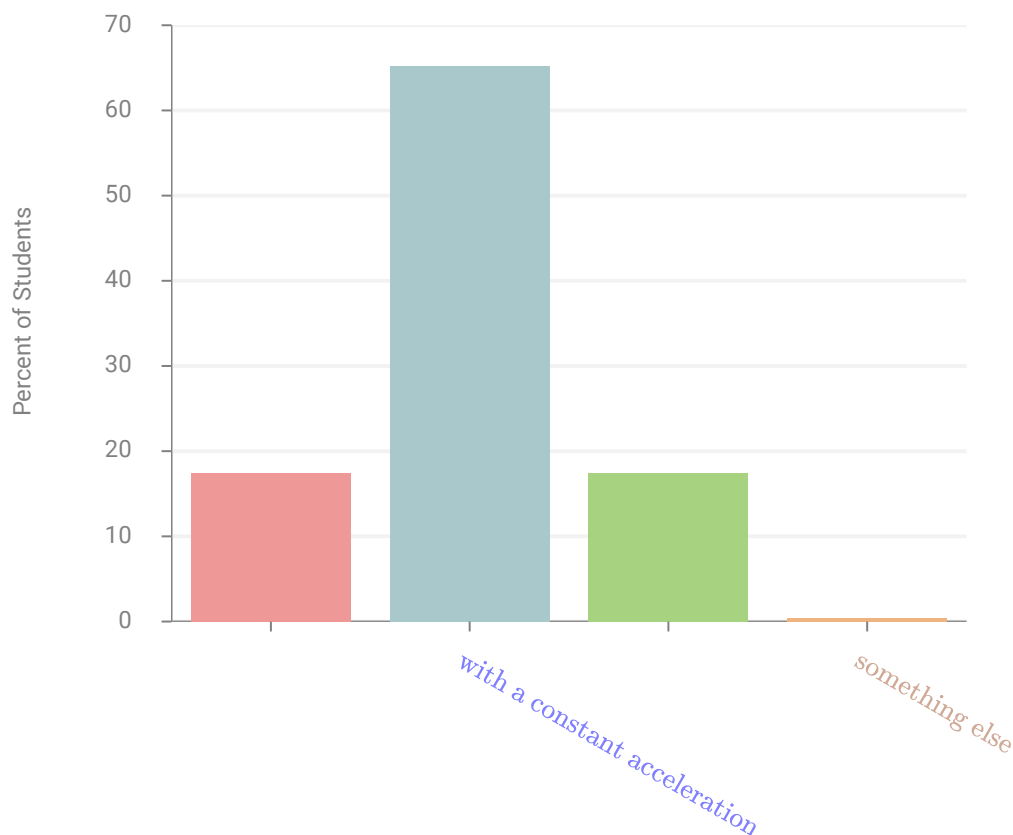
With a constant speed — the speed of gravity

with a constant acceleration

with bigger and bigger acceleration over time

something else

aninconle 2026-09-18 20:43:06



aninconle 2026-09-18 20:43:08

If the speed were constant and an apple fell on your head, it wouldn't matter whether the apple was dropped from one centimeter or one hundred meters—it would hurt the same because it would hit your head at the same speed. Instead, objects get faster and faster as they fall.

aninconle 2026-09-18 20:43:10

In the handout, we studied the model that all objects accelerate towards Earth with the same acceleration. We call the magnitude of that acceleration g .

aninconle 2026-09-18 20:43:13

g varies from place to place on Earth, but is roughly 9.8 m/s^2 .

aninconle 2026-09-18 20:43:15

From Acceleration to Velocity to Distance

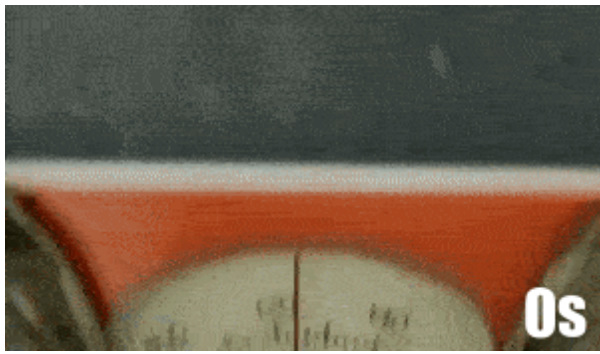
aninconle 2026-09-18 20:43:16

Use a scale and a stopwatch to find the height of an elevator.

aninconle 2026-09-18 20:43:19

Here's a GIF of someone standing on a scale while riding an elevator.

aninconle 2026-09-18 20:43:20



aninconle 2026-09-18 20:43:38

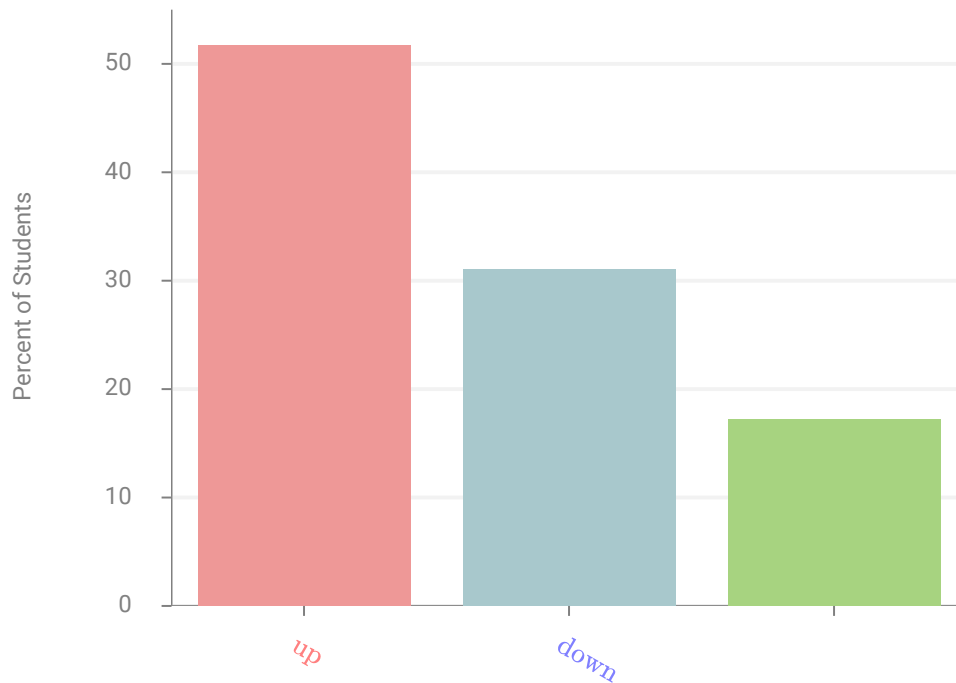
Do you think the person was going up or down?

up

down

no way to say — it would be the same

aninconle 2026-09-18 20:44:25



aninconle 2026-09-18 20:44:29

The reading on the scale goes up at the beginning of the ride, then stays steady, then goes down at the end. This means the person was riding upward.

aninconle 2026-09-18 20:44:35

To understand the connection better, let's think about what acceleration feels like. If you're in a car that suddenly accelerates forward really hard, what do you feel?

VeryDragon2013 2026-09-18 20:45:16

like you're being pushed backwards into the seat

Potatomiaomi 2026-09-18 20:45:16

ur getting pushed into ur seat

leopards 2026-09-18 20:45:16

you feel like you got pushed backwards

Zip-a-Dee-Doo-Dah 2026-09-18 20:45:16

You get pushed backwards against the seat.

apollo2000 2026-09-18 20:45:16

I feel like im flying backward

hw_zasf.xAx 2026-09-18 20:45:16

pushed backwards

Nebula233 2026-09-18 20:45:16

pushed back against seat

Bunnicorn 2026-09-18 20:45:16

you get pushed into your seat

loripachiardi 2026-09-18 20:45:16

pulling back, towards the seat

Grace_174 2026-09-18 20:45:16

I feel pushed back against the seat

aninconle 2026-09-18 20:45:18

You feel pushed backward in your seat, the opposite direction as the acceleration.

aninconle 2026-09-18 20:45:19



aninconle 2026-09-18 20:45:25

After a few seconds of being pushed backwards, the people went back to normal.

aninconle 2026-09-18 20:45:27

When did the people stop feeling like they were pushed back in their seat?

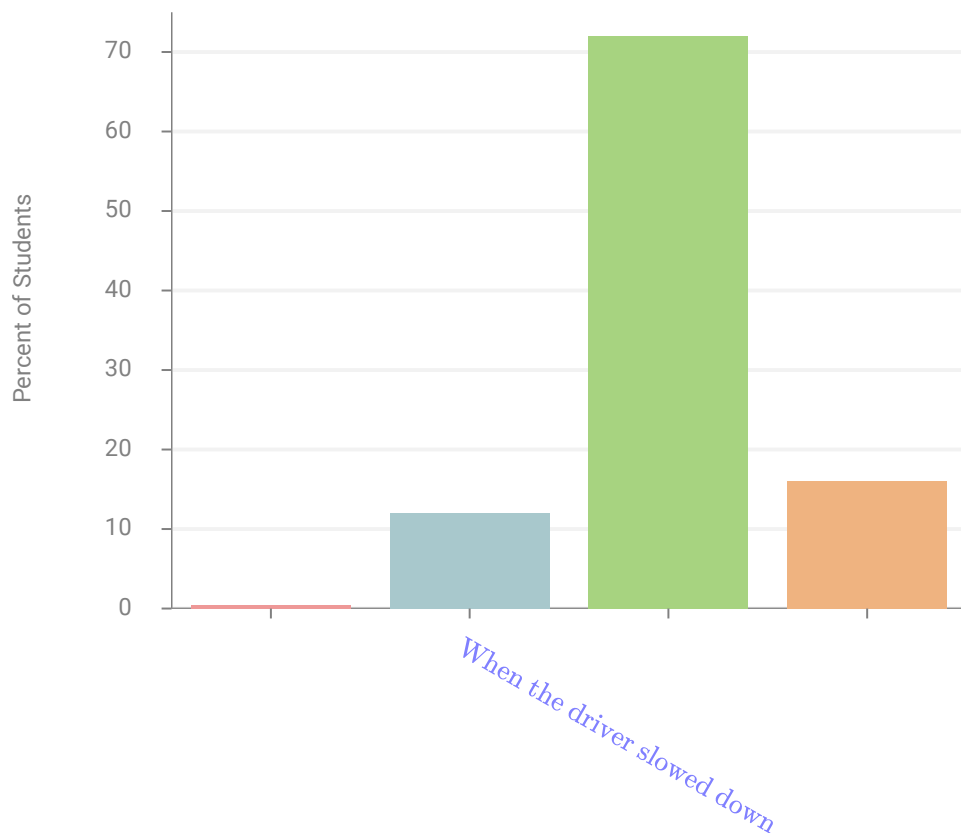
When the driver stopped the car

When the driver slowed down

When the driver continued at a constant maximum velocity

When the driver kept accelerating, but the people adjusted to it

aninconle 2026-09-18 20:46:02



aninconle 2026-09-18 20:46:05

You feel pushed around when there is acceleration, not high speed. The "push" turned off when the driver reached max speed.

aninconle 2026-09-18 20:46:06

What would the passengers have felt if the driver had started slowing down?

syl314 2026-09-18 20:47:12

Pushed forward

Oceanix 2026-09-18 20:47:12

Pushed forward

hw_zasf.xAx 2026-09-18 20:47:12

pushed forwards

leopards 2026-09-18 20:47:12

pushed forwards

Ash_Ketchup13 2026-09-18 20:47:12

forward push

flybirdeee_ 2026-09-18 20:47:12

leaned forward

AN29 2026-09-18 20:47:12

they would have leaned forward

M.Zhang555 2026-09-18 20:47:12

pushed forward

Grace_174 2026-09-18 20:47:12

the momentum forward

kabir_sahil 2026-09-18 20:47:12

Thrown forward

Canada1334 2026-09-18 20:47:12

They would fly forward

Supertomato1919 2026-09-18 20:47:12

a forward force

kanishka.chittoor 2026-09-18 20:47:12

going forward

veryhungrypanda 2026-09-18 20:47:12

going forward

lullipvella 2026-09-18 20:47:12

they were going forwards

Seifr 2026-09-18 20:47:12

Pushed forward.

Nebula233 2026-09-18 20:47:12

pushed forward

VeryDragon2013 2026-09-18 20:47:12

pushed forwards

aninconle 2026-09-18 20:47:13

They'd be pushed forward.

aninconle 2026-09-18 20:47:14



aninconle 2026-09-18 20:47:17

Again, this is the opposite direction of the acceleration, since slowing down is a backward acceleration when the car is going forward.

aninconle 2026-09-18 20:47:20

We can call this "feeling gees." When a car accelerates forward at about 5 m/s^2 , the feeling you get when sitting in the car is the same as if there were a freefall acceleration of 5 m/s^2 backwards on you, in addition to the normal downward one.

aninconle 2026-09-18 20:47:23

The idea that acceleration and gravity feel exactly the same is called the *equivalence principle*. Einstein used it as an important part of his invention of *general relativity*, a theory of gravity.

aninconle 2026-09-18 20:47:24

When you're in an elevator that's just beginning its ascent, what do you feel?

Bunnicorn 2026-09-18 20:48:00

pushed down

apollo2000 2026-09-18 20:48:00

a momentum downwards

syl314 2026-09-18 20:48:00

Pushed down

kanishka.chittoor 2026-09-18 20:48:00

pushed down

TatuShark 2026-09-18 20:48:00

pushed down

aahelic13 2026-09-18 20:48:00

you feel you body going down

Potatomiaomi 2026-09-18 20:48:00

getting pushed upwards

aninconle 2026-09-18 20:48:02

The elevator is accelerating up, so you feel as if there's extra free fall acceleration downward. It's as if gravity were extra strong. If the acceleration is 2 m/s^2 , then you feel about 20% heavier than normal, since $g \approx 10 \text{ m/s}^2$.

aninconle 2026-09-18 20:48:03



aninconle 2026-09-18 20:48:19

The reading on the scale rises from about 124 lbs to 130 lbs, an increase of $\frac{6 \text{ lbs}}{124 \text{ lbs}} = 0.05$. That means the magnitude of the acceleration was about

$$\begin{aligned} a &\approx 0.05 \cdot 10 \text{ m/s}^2 \\ &= 0.5 \text{ m/s}^2. \end{aligned}$$

aninconle 2026-09-18 20:48:29

We now have an acceleration. What other information are we going to need from the GIF to figure out how tall the elevator is?

Ash_Ketchup13 2026-09-18 20:49:27

time

syl314 2026-09-18 20:49:27

The time

flybirdeee_ 2026-09-18 20:49:27

total time

Marshall_Huang 2026-09-18 20:49:27

the time

Oceanix 2026-09-18 20:49:27

time

VeryDragon2013 2026-09-18 20:49:27

The total times

Zip-a-Dee-Doo-Dah 2026-09-18 20:49:27

The total time?

hw_zasf.xAx 2026-09-18 20:49:27

the seconds counted

Grace_174 2026-09-18 20:49:27

the time duration?

apollo2000 2026-09-18 20:49:27

the total time

aninconle 2026-09-18 20:49:30

We'll need to know how long the total ride took and how long the acceleration period was.

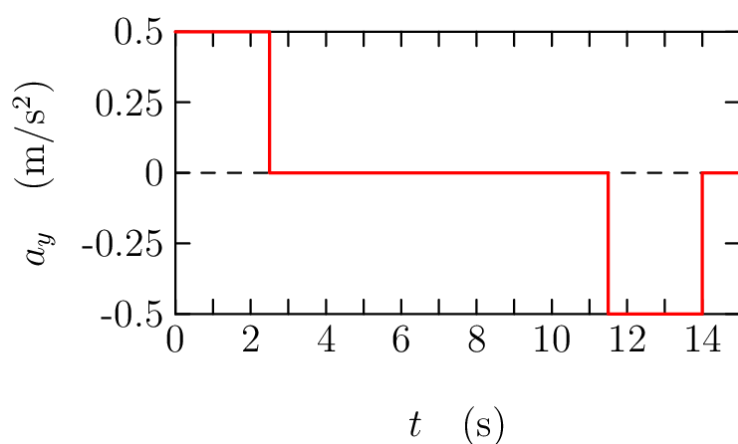
aninconle 2026-09-18 20:49:32

It looks like the acceleration lasts for about 2.5 s at both the start and end of the ride, and that the ride is 14 s long.

aninconle 2026-09-18 20:49:35

Here's a graph that models the acceleration over time, based on what we've said so far:

aninconle 2026-09-18 20:49:36



aninconle 2026-09-18 20:49:40

During which part(s) is the velocity of the elevator changing?

VeryDragon2013 2026-09-18 20:50:23

the first and last part

apollo2000 2026-09-18 20:50:23

0-2 and 12-14

M.Zhang555 2026-09-18 20:50:23

0 to 2.5 and 11.5 to 14

Zip-a-Dee-Doo-Dah 2026-09-18 20:50:23

The beginning and very close to the end.

AN29 2026-09-18 20:50:23

0-2s, 12-14s

flybirdeee_ 2026-09-18 20:50:23

0-2 seconds and 11-14 seconds

kabir_sahil 2026-09-18 20:50:23

0 to 2.5 secs and 11.5 to 14 secs

apollo2000 2026-09-18 20:50:23

0-3 and 11-14

aninconle 2026-09-18 20:50:24

The velocity changes whenever the acceleration isn't zero. That's continuously from 0 s to 2.5 s and again from 11.5 s to 14 s.

aninconle 2026-09-18 20:50:45

AT 0 to 2.5 it's traveling upward at constant velocity. The acceleration is zero, but the elevator is still moving. However, you can't feel velocity, only acceleration, so the person standing on the scale sees just their normal weight during this time.

aninconle 2026-09-18 20:50:51

About how fast does the elevator go during its constant-velocity part of the trip upward?

aninconle 2026-09-18 20:51:51

We can find this by multiplying the positive acceleration by the duration of that acceleration,

$$\begin{aligned} v_{\max} &= (0.5 \text{ m/s}^2) \cdot (2.5 \text{ s}) \\ &= 1.25 \text{ m/s.} \end{aligned}$$

aninconle 2026-09-18 20:51:57

Do you see a way to interpret the quantity $v_{\max} = 1.25 \text{ m/s}$ as some piece of the a versus t plot?

AN29 2026-09-18 20:53:25

is it the area under the curve?

VeryDragon2013 2026-09-18 20:53:25

The area under the acceleration

Seffr 2026-09-18 20:53:25

The area?

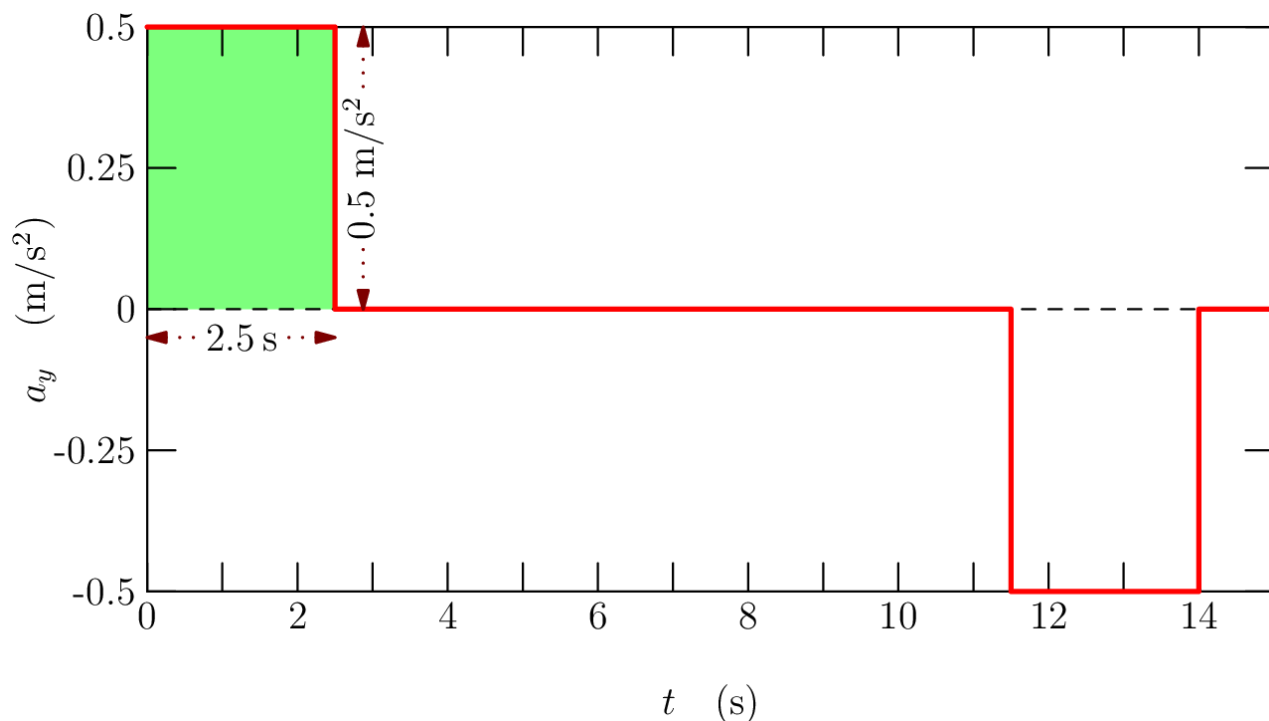
thekitsune 2026-09-18 20:53:27

area

aninconle 2026-09-18 20:53:29

It's this area:

aninconle 2026-09-18 20:53:30



aninconle 2026-09-18 20:53:38

And when the elevator slows down near the top, what happens to the velocity, and how can we see that on the plot?

syl314 2026-09-18 20:55:15

It goes back down to zero, as we can see by the negative area on the other side

leopards 2026-09-18 20:55:15

the area under the negative acceleration part

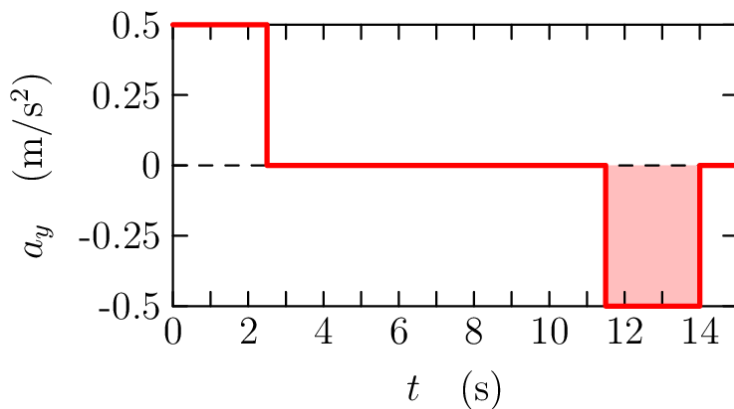
VeryDragon2013 2026-09-18 20:55:15

it decreases, it is the area above the curve below $x=0$

aninconle 2026-09-18 20:55:17

The velocity decreases. We see that because the area under the second hump is beneath the horizontal axis, indicating it's negative.

aninconle 2026-09-18 20:55:18



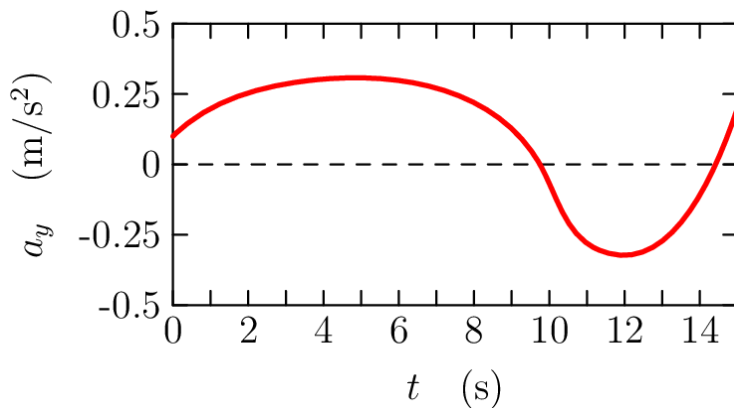
aninconle 2026-09-18 20:55:19

Our conclusion is that the area between a plot of acceleration versus time and the horizontal axis is the change in velocity of an object.

aninconle 2026-09-18 20:55:23

This is true even for complicated acceleration versus time plots. For example, if we had an acceleration versus time plot like this:

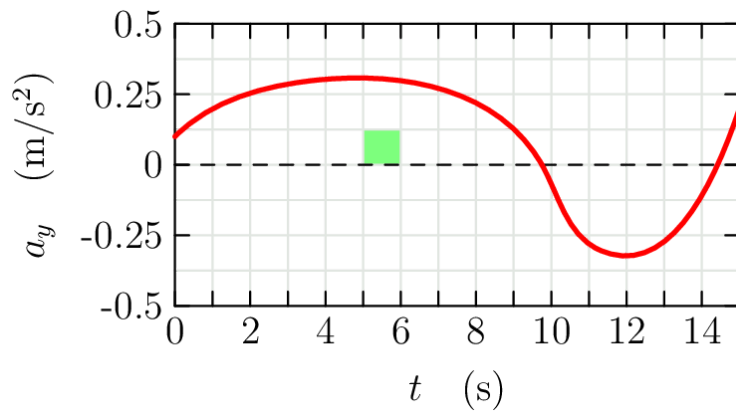
aninconle 2026-09-18 20:55:24



aninconle 2026-09-18 20:55:29

We could put a grid on it to divide it into little rectangles, one of which I've highlighted below:

aninconle 2026-09-18 20:55:30



aninconle 2026-09-18 20:55:33

That little rectangle represents a time of 1 s and an acceleration of 0.125 m/s^2 , so it represents a velocity change of $\Delta v_{y,\text{rectangle}} = 0.125 \text{ m/s}$.

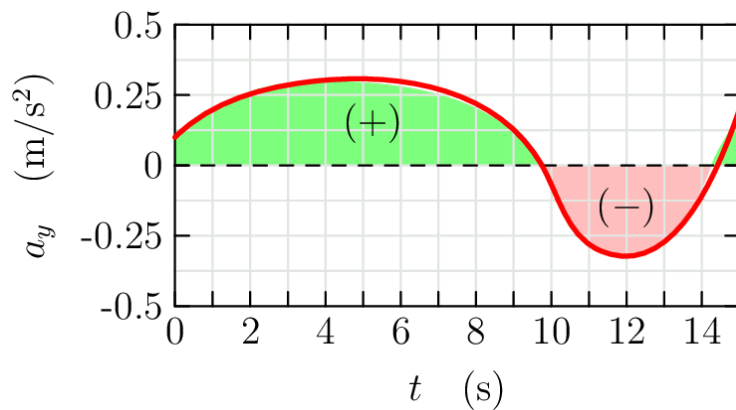
aninconle 2026-09-18 20:55:37

Every rectangle above the t axis (the line $a_y = 0$) and below the curve contributes the same amount to the change in velocity of the object. Every rectangle below $a_y = 0$ and above the curve subtracts $\Delta v_{y,\text{rectangle}}$ from the velocity change of the object. Partial rectangles make a partial contribution.

aninconle 2026-09-18 20:55:41

Adding up all the rectangles, we get the *signed area* between the curve and the horizontal axis. You'll also see this called the "area under the curve."

aninconle 2026-09-18 20:55:43



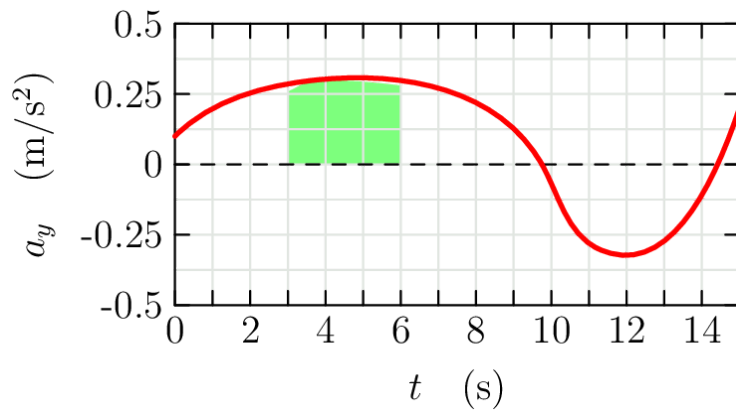
aninconle 2026-09-18 20:55:47

This area tells us the total Δv_y of the object over the time we've plotted the acceleration.

aninconle 2026-09-18 20:55:53

We could also find the signed area over some particular time to find the Δv_y over that time. For example, here's the signed area from 3 s to 6 s.

aninconle 2026-09-18 20:55:55



aninconle 2026-09-18 20:55:58

This area is how much the velocity changes between 3 s and 6 s.

syl314 2026-09-18 20:56:08

Integrals!

Exponent11 2026-09-18 20:56:08

this is calculus

aninconle 2026-09-18 20:56:14

Yep, this is calculus 🤖

aninconle 2026-09-18 20:56:17

About how much does the velocity of the elevator increase from 3 s to 6 s in this plot?

$$\Delta v_y \approx \boxed{} \text{ m/s}$$

leopards 2026-09-18 20:58:09

$$\Delta v_y \approx 0.9 \text{ m/s}$$

Thorongil 2026-09-18 20:58:09

$$\Delta v_y \approx 0.6 \text{ m/s}$$

VeryDragon2013 2026-09-18 20:58:09

$$\Delta v_y \approx 0.94 \text{ m/s}$$

apollo2000 2026-09-18 20:58:09

$$\Delta v_y \approx 0.9 \text{ m/s}$$

TatuShark 2026-09-18 20:58:09

$$\Delta v_y \approx 0.75 \text{ m/s}$$

agoel2010 2026-09-18 20:58:09

$$\Delta v_y \approx .75 \text{ m/s}$$

Grace_174 2026-09-18 20:58:09

$$\Delta v_y \approx 0.75 \text{ m/s}$$

Potatomiaomi 2026-09-18 20:58:09

$$\Delta v_y \approx 0.7 \text{ m/s}$$

Thorongil 2026-09-18 20:58:09

$$\text{oh, oops } \Delta v_y \approx 0.85 \text{ m/s}$$

aninconle 2026-09-18 20:58:11

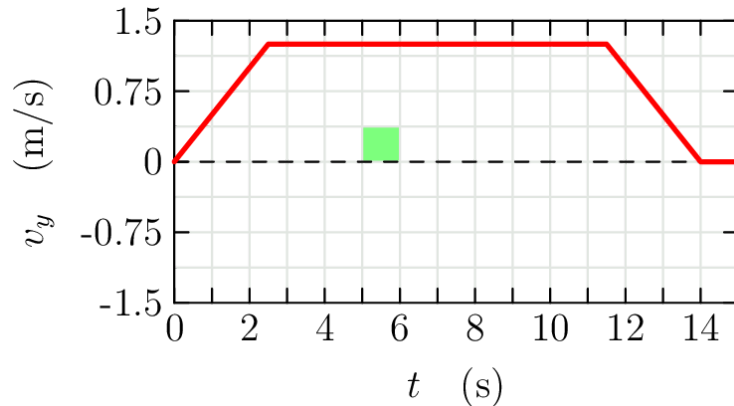
There are six full rectangles and three partial rectangles at top. That's about seven rectangles total, and

$$\begin{aligned}\Delta v_y &\approx 7 \cdot 0.125 \text{ m/s} \\ &= 0.875 \text{ m/s} \\ &\approx 0.9 \text{ m/s}.\end{aligned}$$

aninconle 2026-09-18 20:58:14

If we know the velocity's initial value, we can use the acceleration versus time plot to find the velocity at any time. That lets us generate a velocity versus time curve, like this one for the elevator:

aninconle 2026-09-18 20:58:16



aninconle 2026-09-18 20:58:17

(We're now going back to the elevator GIF, not using the curvy acceleration versus t plot anymore.)

aninconle 2026-09-18 20:58:19

What type of quantity does the area of that little green rectangle represent now?

anisquash 2026-09-18 20:59:20

Displacement

M.Zhang555 2026-09-18 20:59:20

i think displacement

apollo2000 2026-09-18 20:59:20

the change in position

aninconle 2026-09-18 20:59:22

The height of the rectangle is a velocity and the width is a time. Multiplying those to get an area, we multiply a velocity by a time, getting a displacement.

aninconle 2026-09-18 20:59:24

How much displacement does each rectangle represent?

$$\Delta y_{\text{rectangle}} \approx \boxed{} \text{ m}$$

Bunicorn 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

Nebula233 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

flybirdeee_ 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

apollo2000 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

anisquash 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

starkmh002 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

Thorongil 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

kabir_sahil 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

Oceanix 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

Grace_174 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx .375 \text{ m}$$

Potatomiaomi 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

TatuShark 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

Ash_Ketchup13 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

Seifr 2026-09-18 21:00:44

$$\Delta y_{\text{rectangle}} \approx 0.375 \text{ m}$$

aninconle 2026-09-18 21:00:46

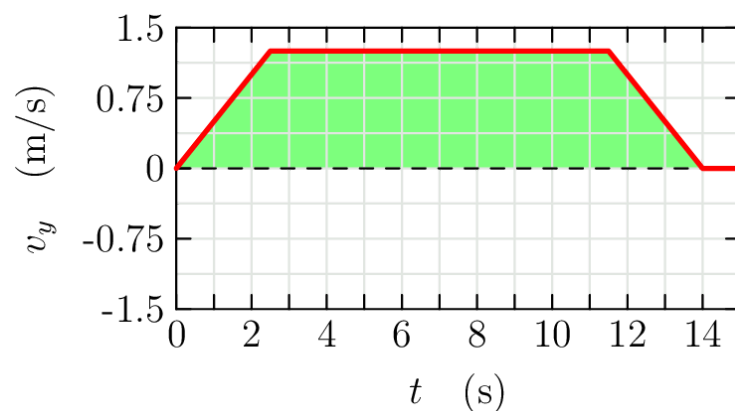
The height of a rectangle is $\frac{1}{2} \cdot 0.75 \text{ m/s} = 0.375 \text{ m/s}$. The displacement is

$$\begin{aligned}\Delta y_{\text{rectangle}} &\approx 0.375 \text{ m/s} \cdot 1 \text{ s} \\ &= 0.375 \text{ m}.\end{aligned}$$

aninconle 2026-09-18 21:00:51

If we add up all the rectangles between the curve and the horizontal axis, we get the total displacement of the elevator, represented by the total area between the curve and the horizontal axis.

aninconle 2026-09-18 21:00:52



aninconle 2026-09-18 21:00:53

About how far did the elevator rise?

$$\Delta y \approx \text{ } \text{m}$$

Oceanix 2026-09-18 21:02:10

$$\Delta y \approx 13 \text{ m}$$

Ash_Ketchup13 2026-09-18 21:02:10

$$\Delta y \approx 13.5 \text{ m}$$

leopards 2026-09-18 21:02:10

$$\Delta y \approx 14 \text{ m}$$

VeryDragon2013 2026-09-18 21:02:10

$$\Delta y \approx 15 \text{ m}$$

Thorongil 2026-09-18 21:02:10

$$\Delta y \approx 14.375 \text{ m}$$

apollo2000 2026-09-18 21:02:10

sorry, i meant 13.8 m

M.Zhang555 2026-09-18 21:02:10

$$\Delta y \approx 14.375 \text{ m}$$

aninconle 2026-09-18 21:02:12

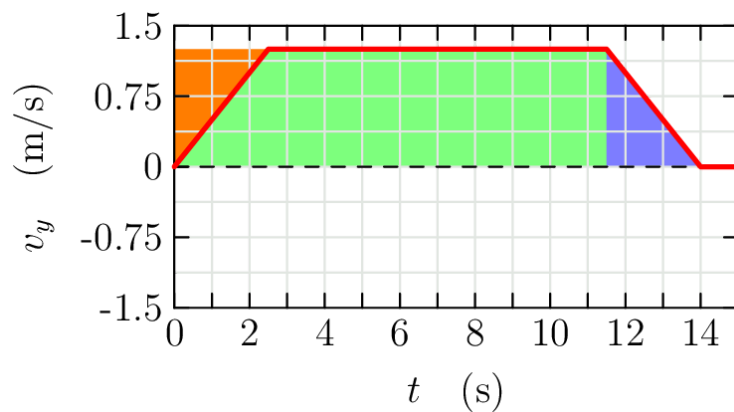
We could count the rectangles. Including partials, there are roughly 38, giving

$$\begin{aligned}\Delta y &\approx 38 \cdot 0.375 \text{ m} \\ &= 14.25 \text{ m}.\end{aligned}$$

aninconle 2026-09-18 21:02:13

Alternatively, let's take the area in blue below (the area from 11.5 s to 14 s) and move it over to the area in orange (an area above the curve right at the beginning).

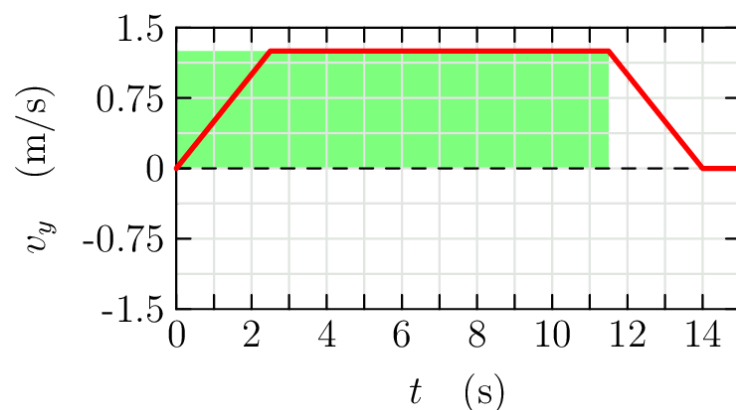
aninconle 2026-09-18 21:02:15



aninconle 2026-09-18 21:02:16

This way of chopping up the area between the velocity curve and the horizontal axis gives us a single rectangle of width 11.5 s and height 1.25 m/s.

aninconle 2026-09-18 21:02:17



aninconle 2026-09-18 21:02:18

The displacement of the elevator is

$$\begin{aligned}\Delta y &\approx 1.25 \text{ m/s} \cdot 11.5 \text{ s} \\ &= 14 \text{ m.}\end{aligned}$$

aninconle 2026-09-18 21:02:20

Our model was also a bit crude, using perfectly constant acceleration. Our acceleration measurements had some significant uncertainty, too. We shouldn't expect this result to be extremely accurate.

aninconle 2026-09-18 21:02:26

Still, one floor in a building is usually around 4.3 m in height.

aninconle 2026-09-18 21:02:28

We can calculate

$$\frac{14 \text{ m}}{4.3 \text{ m/floor}} = 3.25 \text{ floor.}$$

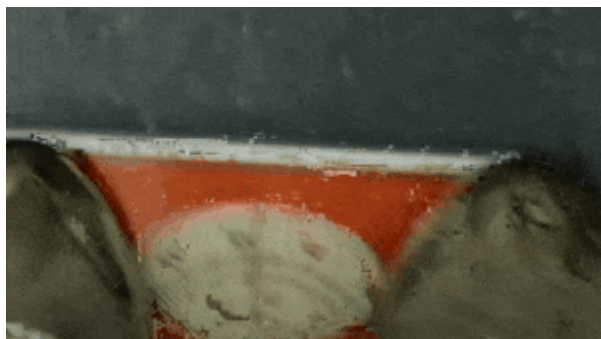
aninconle 2026-09-18 21:02:30

If we round that to three floors, the rider went from the first floor to the fourth.

aninconle 2026-09-18 21:02:32

That agrees with what the source video says:

aninconle 2026-09-18 21:02:33



aninconle 2026-09-18 21:02:38

The next time your elevator stops telling you what floor you're on, you don't have to give up and *scale* the building yourself—acceleration will give you a *lift*!

apollo2000 2026-09-18 21:03:02

hahaha

flybirdeee_ 2026-09-18 21:03:02

haha

VeryDragon2013 2026-09-18 21:03:02

oh man the puns....

Canada1334 2026-09-18 21:03:02

haha

Supertomato1919 2026-09-18 21:03:02

hahaha

syl314 2026-09-18 21:03:02

Lol

pandafrog 2026-09-18 21:03:09

Cool!

Supertomato1919 2026-09-18 21:03:09

ha

guacataj 2026-09-18 21:03:09

lol!

Ash_Ketchup13 2026-09-18 21:03:09

yay

aninconle 2026-09-18 21:03:10

Alright! We are out of time. That leaves us one last thing to do ...

guacataj 2026-09-18 21:03:26

SUMMARY

pandafrog 2026-09-18 21:03:26

$M + A + R + Y$

flybirdeee_ 2026-09-18 21:03:26

sumary

Zip-a-Dee-Doo-Dah 2026-09-18 21:03:26

$\sum$ MARY

Supertomato1919 2026-09-18 21:03:32

Say goodbye

Nebula233 2026-09-18 21:03:32

summarayy?

syl314 2026-09-18 21:03:32

Summary!

aninconle 2026-09-18 21:03:39

POTATO

aninconle 2026-09-18 21:03:41

Any time an object's velocity is changing, we say it has non-zero *acceleration*.

aninconle 2026-09-18 21:03:42

Acceleration can be positive or negative. The sign of acceleration tells you whether a velocity versus time plot slopes up or down.

aninconle 2026-09-18 21:03:43

If we know an acceleration, we can use it to calculate how much a velocity changes. The change in velocity is the signed area between an acceleration versus time plot and the horizontal axis.

aninconle 2026-09-18 21:03:45

The displacement of an object is the signed area between the horizontal axis and a velocity versus time plot.

aninconle 2026-09-18 21:03:47

By working from acceleration to velocity to displacement, knowledge of acceleration over time, plus initial velocity, lets us find an object's displacement.

aninconle 2026-09-18 21:03:49

You can feel acceleration. When something starts accelerating you, it feels as if you're being pushed in the opposite direction, exactly as if gravitational acceleration had changed. This is called the *equivalence principle*. You cannot feel velocity in a similar way.

aninconle 2026-09-18 21:03:50

Credits

bowling ball and feather: BBC, "Brian Cox visits the world's biggest vacuum | Human Universe" <https://www.youtube.com/watch?v=E43-CfukEgs>

scale in elevator: Mr Guay Hansen, "Is it mass or weight? Using a bathroom scale in an elevator" <https://www.youtube.com/watch?v=E2rI4m7N9-Q>

free fall experiment: phyphox, "Smartphone-Experiment: Free fall (en)" https://www.youtube.com/watch?v=zRGh9_a1J7s

car acceleration pushed backwards: Scott Hilse, "Tesla Model S P100D Ludicrous+ Launch Reactions," <https://www.youtube.com/watch?v=I5mZ1IK7SuU>

car stopping acceleration: Channel 9, "The biggest fails from Driving Test | Driving Test 2020," <https://www.youtube.com/watch?v=l7afy7SGZxE>

vertical jump: The Oregonian, "Watch Josh Imatorbhebhe's 47.1 inch vertical jump," <https://www.youtube.com/watch?v=lgkCxnSHV7w>

aninconle 2026-09-18 21:03:52

Have fun!



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